HCC Survival Analysis

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1 Overview

This R Markdown document restructures the original HCC survival analysis script into a report format so the code, tables, model output, and plots can all be shown together after knitting.

2 Load required packages

```r
# install.packages("labelled")
# install.packages("gmodels")
# install.packages("survminer")
# install.packages("survRM2")

library(haven)
library(dplyr)
library(survival)
library(survminer)
library(labelled)
library(readr)
library(scales)
library(forcats)
library(gmodels)
library(stringr)
library(tibble)
library(lubridate)
```

3 Import dataset

```r
file_path <- "C:/Users/saeeda1/OneDrive - UMass Chan Medical School/Documents/independent study/cancer p

chart_data <- read_sas(file_path)
glimpse(chart_data)
```

```
## Rows: 89
## Columns: 59
## $ unique_id          <dbl> 2, 3, 7, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20~
## $ study_id           <dbl> 2, 3, 6, 8, 9, 10, 11, 11, 12, 13, 14, 15, 16, 18,~
## $ unique             <dbl> 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0,~
## $ age_first_SBRT     <dbl> 59, 61, 77, 64, 36, 51, 63, 63, 51, 64, 66, 65, 79~
## $ Milan_criteria     <dbl> 1, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1,~
## $ pvtt               <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,~
## $ etiology_1         <dbl> 2, 1, 5, 2, 3, 2, 2, 2, 1, 8, 2, 1, 5, 1, 3, 3, 3,~
## $ etiology_2         <dbl> 1, NA, NA, NA, 1, 1, 1, 1, NA, NA, NA, 2, NA, 2, N~
## $ ascites            <dbl> 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0,~
## $ listed             <chr> "1", "1", "0", "0", "0", "0", "0", "0", "0", "1", ~
## $ n_tumor_LDT1       <dbl> 2, 2, 1, 2, 6, 1, 3, 3, 2, 1, 2, 3, 1, 1, 3, 2, 3,~
## $ date_LTD1          <date> 2012-05-08, 2011-04-07, 2012-09-05, 2012-07-01, 2~
## $ n_hcc_tx_LTD1      <dbl> 2, 2, 1, 2, 6, 1, 3, 3, 2, 1, 2, 3, 1, 1, 3, 2, 3,~
## $ type_LDT1          <chr> "TACE", "TACE", "TACE", "TACE", "TACE", "TACE", "T~
## $ size_tx_lesion     <dbl> 3.2, 5.5, 3.7, 4.5, 1.1, 4.1, 2.0, 2.0, 2.8, 2.7, ~
## $ date_LDT2          <date> NA, 2012-06-25, NA, NA, 2012-11-20, NA, NA, NA, 2~
## $ type_LDT2          <chr> "SBRT", "RFA", "SBRT", "SBRT", "TACE", "0", "SBRT"~
## $ date_LDT3          <date> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ type_LDT3          <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", ~
## $ date_LDT4          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ type_LDT4          <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", ~
## $ prior_LDT          <dbl> 1, 2, 1, 1, 2, 1, 1, 1, 2, 1, 1, 2, 1, 2, 1, 1, 1,~
## $ date_last_LDT      <date> 2012-05-08, 2012-06-25, 2012-09-05, 2012-07-01, 2~
```

```
## $ last_LDT            <chr> "TACE", "RFA", "TACE", "TACE", "TACE", "TACE", "TA~
## $ Desc_Salv_Consol    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ n_hcc__time_sbrt    <dbl> 2, 2, 1, 2, 5, 1, 3, 3, 2, 1, 1, 3, 1, 1, 3, 3, 3,~
## $ n_hcc_tx_sbrt       <dbl> 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 1, 3, 3, 3,~
## $ hcc_size_sbrt       <dbl> 3.2, 5.5, 3.7, 4.5, 1.1, 4.1, 2.0, 2.0, 2.8, 2.7, ~
## $ cp_prior_group      <dbl> 1, 1, 1, 2, 1, 1, 2, 2, 1, 2, 1, 2, 1, 1, 1, 1, 1,~
## $ afp_prior_sbrt      <dbl> 6.15, 14.90, 2.32, 27.70, 442.00, 9.24, 10.70, 10.~
## $ td                  <dbl> 50, 54, 54, 30, 50, 45, 35, 35, 55, 40, 60, 40, 50~
## $ n_fx                <dbl> 5, 3, 3, 5, 3, 3, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,~
## $ eqd2                <dbl> 83.33333, 126.00000, 126.00000, 40.00000, 111.1111~
## $ sbrt_start          <date> 2012-08-27, 2012-08-29, 2012-11-28, 2013-02-13, 2~
## $ sbrt_end            <date> 2012-09-10, 2012-09-12, 2012-12-04, 2013-02-22, 2~
## $ gtv                 <chr> "16.3", "5.1", "57.9", "47.2", "6.2", "54", "45.1"~
## $ itv                 <chr> "23.7", "10.7", "90.9", "64.3", "6.8", "141.2", "1~
## $ ptv                 <dbl> 32.9, 17.6, 99.6, 86.8, 16.1, 171.1, 200.5, 200.5,~
## $ mld                 <dbl> 8.0, 4.4, 9.0, 9.1, 2.9, 11.1, 4.5, 4.5, 7.7, 8.9,~
## $ afp_1_mo_after      <chr> "1.09", "17.1", "2.41", "16.4", "4.25", "18.6", "6~
## $ cp_comp_base_6_mo   <dbl> 2, 2, 2, 3, 3, 2, 0, 0, 2, 3, 2, 1, 1, 1, 1, 1, 1,~
## $ cp_miss_data        <dbl> 0, 0, 0, 0, 0, 0, 4, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ gd_3_tx             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,~
## $ elpd                <dbl> 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0,~
## $ elpd_date           <date> NA, NA, NA, 2013-07-17, NA, 2016-03-21, NA, NA, N~
## $ dm                  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,~
## $ dm_date             <date> 1899-12-30, 1899-12-30, 1899-12-30, 1899-12-30, 1~
## $ lc_censor           <date> 2013-02-15, 2018-06-13, 2013-06-04, 2013-12-01, 2~
## $ lc_time             <dbl> 5.166667, 69.033333, 6.000000, 9.300000, 29.333333~
## $ lc_censor_reason    <chr> "OLT", "OLT", "FU", "died", "died", "died", "died"~
## $ resp                <chr> "pPR", "pCR", "cCR", "cPR", "cCR", "cCR", "cPR", "~
## $ radio_resp          <chr> "cPR", "cCR", "cCR", "cPR", "cCR", "cCR", "cPR", "~
## $ path_resp           <chr> "pPR", "pCR", "NA", "NA", "NA", "NA", "NA", "NA", ~
## $ transplant          <dbl> 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1,~
## $ os_censor           <date> 2014-08-03, 2023-09-19, 2014-12-23, 2013-12-01, 2~
## $ os_censor_reason    <chr> "died", "FU", "FU", "died", "died", "died", "died"~
## $ os_cmt_salv         <dbl> 26.833333, 149.400000, 27.600000, 17.000000, 36.56~
## $ residual_reccurence <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1,~
## $ cmt_diag_test       <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,~
```

```
hcc_data <- read_sas(file_path)
```

4 Data preparation

4.1 Rename selected variables

```
chart_data <- hcc_data %>%
  mutate(
    residual_recurrence = residual_reccurence,
    date_ldt1 = date_LTD1,
    n_hcc_tx_ldt1 = n_hcc_tx_LTD1
  ) %>%
  select(-residual_reccurence, -date_LTD1, -n_hcc_tx_LTD1)
```

4.2 Apply variable labels

```
var_labels <- c(
  unique_id = "Unique ID",
  study_id = "Study ID",
  unique = "Unique Patient",
  age_first_SBRT = "Age first SBRT",
  Milan_criteria = "Inside Milan",
  pvtt = "PVTT",
  etiology_1 = "Etiology 1",
  etiology_2 = "Etiology 2",
  ascites = "Ascites",
  listed = "Listed",
  n_tumor_LDT1 = "No. Tumors LDT1",
  date_LTD1 = "Date LDT1",
  n_hcc_tx_LTD1 = "No. HCC tumor tx LDT1",
  type_LDT1 = "Type LDT1",
  size_tx_lesion = "Size Tx Lesion",
  date_LDT2 = "Date LDT2",
  type_LDT2 = "Type LDT2",
  date_LDT3 = "Date LDT3",
  type_LDT3 = "Type LDT3",
  date_LDT4 = "Date LDT4",
  type_LDT4 = "Type LDT4",
  prior_LDT = "No. prior LDTs",
  date_last_LDT = "Date last LDT",
  last_LDT = "Last LDT",
  Desc_Salv_Consol = "SBRT Description",
  n_hcc__time_sbrt = "No. HCC tumors at SBRT",
  n_hcc_tx_sbrt = "No. HCC tumors treated SBRT",
  hcc_size_sbrt = "Size of SBRT treated lesion",
  cp_prior_group = "CP Group",
  afp_prior_sbrt = "AFP Prior to SBRT",
  td = "TD",
  n_fx = "No. Fx",
  eqd2 = "EQD2",
  sbrt_start = "SBRT Start",
  sbrt_end = "SBRT End",
  gtv = "GTV Vol",
  itv = "ITV Vol",
  ptv = "PTV Vol",
  mld = "MLD",
  afp_1_mo_after = "AFP 1 mo after SBRT",
  cp_comp_base_6_mo = "CP comparision baseline-6 months",
  cp_miss_data = "CP comparision missing data",
  gd_3_tx = "Gr3+ tox",
  elpd = "ELPD",
  elpd_date = "Date ELPD",
  dm = "DM",
  dm_date = "Date DM",
  lc_censor = "LC Censor Date (after SBRT)",
  lc_time = "LC Time (days)",
  lc_censor_reason = "LC Censor Reason",
```

```
  resp = "Response to SBRT at censor",
  radio_resp = "Radiographic Respose",
  path_resp = "Path response",
  transplant = "Transplant",
  os_censor = "OS Censor Date",
  os_censor_reason = "OS Censor Reason",
  os_cmt_salv = "OS (CMT/salv (T0 date LDT1)",
  residual_reccurence = "Salvage only",
  cmt_diag_test = "CMT Diagnostic Test before SBRT"
)

chart_data <- set_variable_labels(hcc_data, !!!var_labels)
```

4.3 Optional CSV export

```
write_csv(chart_data, "C:/Users/saeeda1/OneDrive - UMass Chan Medical School/Documents/independent study
```

4.4 Recode factors

```
chart_data <- chart_data %>%
  mutate(
    Milan_criteria = factor(Milan_criteria, levels = c(0, 1), labels = c("No", "Yes")),
    pvtt = factor(pvtt, levels = c(0, 1), labels = c("No", "Yes")),
    ascites = factor(ascites, levels = c(0, 1), labels = c("No", "Yes")),
    listed = factor(listed, levels = c("0", "1"), labels = c("No", "Yes")),
    elpd = factor(elpd, levels = c(0, 1), labels = c("No", "Yes")),
    dm = factor(dm, levels = c(0, 1), labels = c("No", "Yes")),
    transplant = factor(transplant, levels = c(0, 1), labels = c("No", "Yes")),
    Desc_Salv_Consol = factor(Desc_Salv_Consol, levels = c(1, 2), labels = c("Salvage", "Consolidate/CMV
    cp_prior_group = factor(cp_prior_group, levels = c(1, 2), labels = c("A", "B/C")),
    cp_comp_base_6_mo = factor(cp_comp_base_6_mo, levels = c(0, 1, 2, 3),
                               labels = c("Missing", "Progression", "Stable", "Improve")),
    cp_miss_data = factor(cp_miss_data, levels = c(4, 5, 6),
                          labels = c("Deceased", "Transplant", "Short F/U")),
    gd_3_tx = factor(gd_3_tx, levels = c(0, 1, 3), labels = c("No", "Yes", "Unknown")),
    residual_reccurence = factor(residual_reccurence, levels = c(1, 2, 3),
                                 labels = c("Residual", "Recurrence", "Not included")),
    cmt_diag_test = factor(cmt_diag_test, levels = c(0, 1, 3),
                           labels = c("No", "Yes", "Not included")),
    etiology_1 = factor(etiology_1, levels = 1:9,
                        labels = c("HCV", "ETOH", "HBV", "IC", "NASH", "Hemo", "Autoimmune", "PBC", "Ot
    etiology_2 = factor(etiology_2, levels = 1:9,
                        labels = c("HCV", "ETOH", "HBV", "IC", "NASH", "Hemo", "Autoimmune", "PBC", "Ot
  )
```

4.5 Create first-record-per-patient dataset

```
studyid_wise <- chart_data %>%
  arrange(study_id) %>%
  group_by(study_id) %>%
  slice(1) %>%
  ungroup()
```

5 Descriptive analyses

5.1 Group counts

```
CrossTable(studyid_wise$Desc_Salv_Consol, chisq = TRUE)
```

```
##
##
##    Cell Contents
## |-------------------------|
## |                       N |
## |         N / Table Total |
## |-------------------------|
##
##
## Total Observations in Table:  77
##
##
##                  |         Salvage | Consolidate/CMV |
##                  |-----------------|-----------------|
##                  |              49 |              28 |
##                  |           0.636 |           0.364 |
##                  |-----------------|-----------------|
##
##
##
##
```

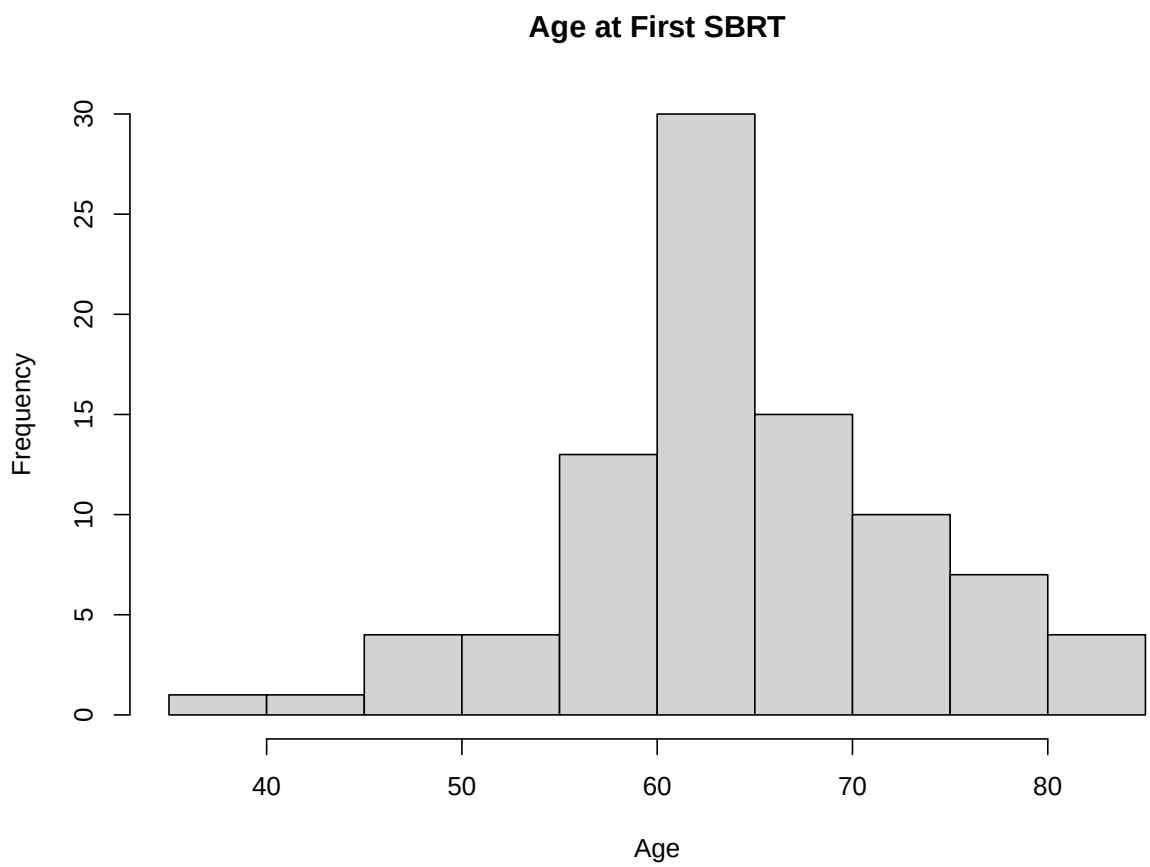
```
CrossTable(chart_data$Desc_Salv_Consol, chisq = TRUE)
```

```
##
##
##    Cell Contents
## |-------------------------|
## |                       N |
## |         N / Table Total |
## |-------------------------|
##
##
## Total Observations in Table:  89
```

```
##
##
##                  |         Salvage | Consolidate/CMV |
##                  |-----------------|-----------------|
##                  |              57 |              32 |
##                  |           0.640 |           0.360 |
##                  |-----------------|-----------------|
##
##
##
##
```

5.2 Age at first SBRT

```r
hist(chart_data$age_first_SBRT, breaks = 10, main = "Age at First SBRT", xlab = "Age")
```




```r
shapiro.test(chart_data$age_first_SBRT)
```

```
##
##  Shapiro-Wilk normality test
##
```

```
## data:  chart_data$age_first_SBRT
## W = 0.9698, p-value = 0.03589
```

```r
summary(chart_data$age_first_SBRT)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   36.00   60.00   63.00   64.33   69.00   85.00
```

```r
sd(chart_data$age_first_SBRT)
```

```
## [1] 9.195818
```

```r
studyid_wise %>%
  group_by(Desc_Salv_Consol) %>%
  summarise(
    median_age = median(age_first_SBRT, na.rm = TRUE),
    Q1 = quantile(age_first_SBRT, 0.25, na.rm = TRUE),
    Q3 = quantile(age_first_SBRT, 0.75, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 4
##   Desc_Salv_Consol median_age    Q1    Q3
##   <fct>                 <dbl> <dbl> <dbl>
## 1 Salvage                  65  61    74
## 2 Consolidate/CMV          64  59.8  68.2
```

```r
t.test(age_first_SBRT ~ Desc_Salv_Consol, data = studyid_wise)
```

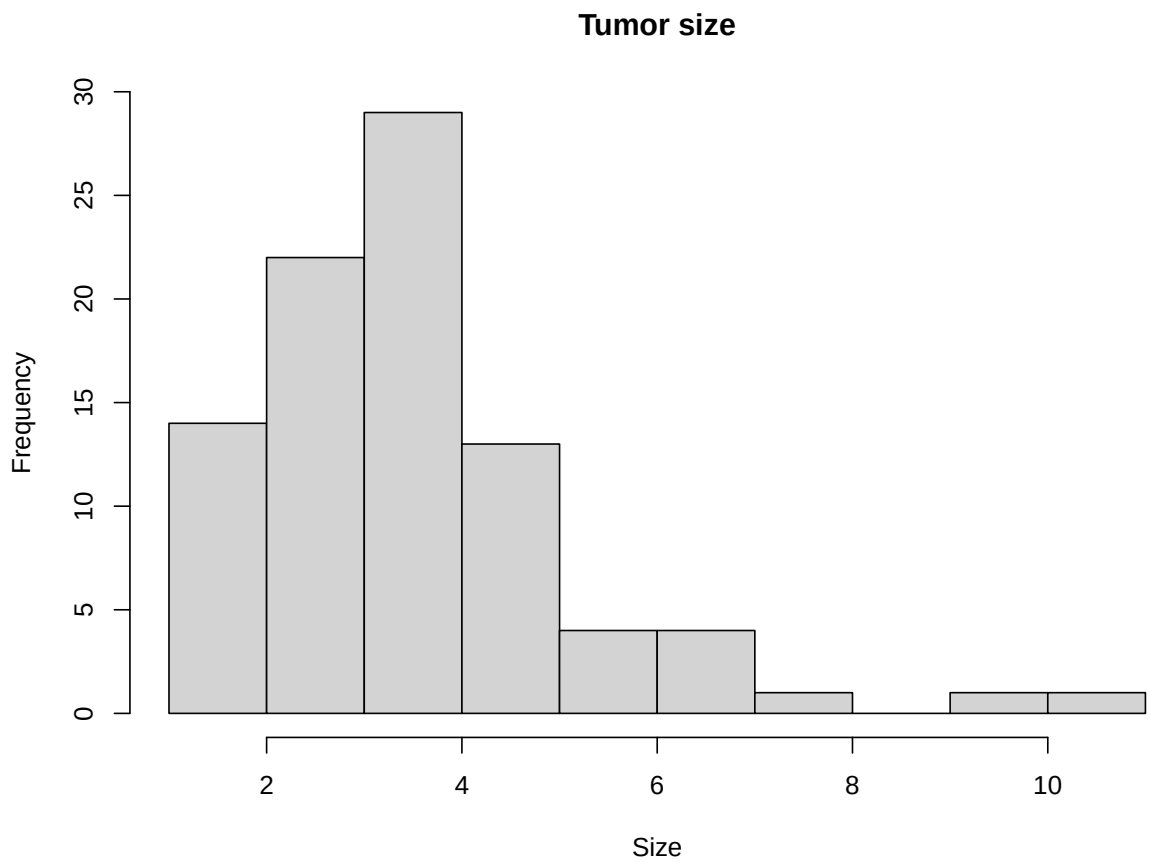
```
##
##  Welch Two Sample t-test
##
## data:  age_first_SBRT by Desc_Salv_Consol
## t = 0.75843, df = 60.73, p-value = 0.4511
## alternative hypothesis: true difference in means between group Salvage and group Consolidate/CMV is n
## 95 percent confidence interval:
##  -2.739063  6.086002
## sample estimates:
##         mean in group Salvage mean in group Consolidate/CMV
##                      65.53061                      63.85714
```

```r
wilcox.test(age_first_SBRT ~ Desc_Salv_Consol, data = studyid_wise, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  age_first_SBRT by Desc_Salv_Consol
## W = 763, p-value = 0.4173
## alternative hypothesis: true location shift is not equal to 0
```

5.3 Tumor size at SBRT

```
hist(chart_data$hcc_size_sbrt, breaks = 10, main = "Tumor size", xlab = "Size")
```




```
shapiro.test(chart_data$hcc_size_sbrt)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  chart_data$hcc_size_sbrt
## W = 0.87719, p-value = 4.983e-07
```

```
chart_data %>%
  group_by(Desc_Salv_Consol) %>%
  summarise(
    median_size = median(hcc_size_sbrt, na.rm = TRUE),
    Q1 = quantile(hcc_size_sbrt, 0.25, na.rm = TRUE),
    Q3 = quantile(hcc_size_sbrt, 0.75, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 4
##   Desc_Salv_Consol median_size    Q1    Q3
##   <fct>                  <dbl> <dbl> <dbl>
## 1 Salvage                 3.2   2.1    4.3
## 2 Consolidate/CMV         3.35  2.98   4
```

```r
t.test(hcc_size_sbrt ~ Desc_Salv_Consol, data = chart_data)
```

```
##
##  Welch Two Sample t-test
##
## data:  hcc_size_sbrt by Desc_Salv_Consol
## t = 0.40728, df = 84.781, p-value = 0.6848
## alternative hypothesis: true difference in means between group Salvage and group Consolidate/CMV is n
## 95 percent confidence interval:
##  -0.5063208  0.7671760
## sample estimates:
##         mean in group Salvage mean in group Consolidate/CMV
##                      3.621053                      3.490625
```

```r
wilcox.test(hcc_size_sbrt ~ Desc_Salv_Consol, data = studyid_wise, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  hcc_size_sbrt by Desc_Salv_Consol
## W = 665.5, p-value = 0.8322
## alternative hypothesis: true location shift is not equal to 0
```

5.4 Selected categorical comparisons

```r
table1 <- table(studyid_wise$etiology_1, studyid_wise$Desc_Salv_Consol)
chisq.test(table1)
```

```
##
##  Pearson's Chi-squared test
##
## data:  table1
## X-squared = 5.2746, df = 8, p-value = 0.7279
```

```r
table(studyid_wise$etiology_2, studyid_wise$Desc_Salv_Consol)
```

```
##
##              Salvage Consolidate/CMV
##   HCV             10               2
##   ETOH             4               2
##   HBV              0               1
##   IC               0               0
##   NASH             0               0
```

```
##   Hemo             0               0
##   Autoimmune       0               0
##   PBC              0               0
##   Other            0               0
```

```r
fisher.test(table(studyid_wise$pvtt, studyid_wise$Desc_Salv_Consol))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(studyid_wise$pvtt, studyid_wise$Desc_Salv_Consol)
## p-value = 0.1523
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.000000 1.862356
## sample estimates:
## odds ratio
##          0
```

```r
chisq.test(table(chart_data$Milan_criteria, chart_data$Desc_Salv_Consol))
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(chart_data$Milan_criteria, chart_data$Desc_Salv_Consol)
## X-squared = 0.55594, df = 1, p-value = 0.4559
```

```r
chisq.test(table(studyid_wise$listed, studyid_wise$Desc_Salv_Consol))
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(studyid_wise$listed, studyid_wise$Desc_Salv_Consol)
## X-squared = 6.9323, df = 1, p-value = 0.008465
```

```r
chisq.test(table(studyid_wise$transplant, studyid_wise$Desc_Salv_Consol))
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(studyid_wise$transplant, studyid_wise$Desc_Salv_Consol)
## X-squared = 4.5839, df = 1, p-value = 0.03227
```

```r
chisq.test(table(studyid_wise$cp_prior_group, studyid_wise$Desc_Salv_Consol))
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(studyid_wise$cp_prior_group, studyid_wise$Desc_Salv_Consol)
## X-squared = 0.50822, df = 1, p-value = 0.4759
```

```r
fisher.test(table(studyid_wise$cp_comp_base_6_mo, studyid_wise$Desc_Salv_Consol))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(studyid_wise$cp_comp_base_6_mo, studyid_wise$Desc_Salv_Consol)
## p-value = 0.334
## alternative hypothesis: two.sided
```

```r
fisher.test(table(studyid_wise$gd_3_tx, studyid_wise$Desc_Salv_Consol))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(studyid_wise$gd_3_tx, studyid_wise$Desc_Salv_Consol)
## p-value = 0.7064
## alternative hypothesis: two.sided
```

5.5 Continuous summary table

```r
chart_data <- chart_data %>%
  mutate(gtv = as.numeric(gtv))

chart_data %>%
  summarise(
    median_gtv = median(gtv, na.rm = TRUE),
    Q1_gtv = quantile(gtv, 0.25, na.rm = TRUE),
    Q3_gtv = quantile(gtv, 0.75, na.rm = TRUE),
    median_ptv = median(ptv, na.rm = TRUE),
    median_td = median(td, na.rm = TRUE),
    median_n_fx = median(n_fx, na.rm = TRUE),
    median_eqd2 = median(eqd2, na.rm = TRUE),
    median_mld = median(mld, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 8
##   median_gtv Q1_gtv Q3_gtv median_ptv median_td median_n_fx median_eqd2
##        <dbl>  <dbl>  <dbl>      <dbl>     <dbl>       <dbl>       <dbl>
## 1       25.6   12.5   44.5       86.8      47.5           5        77.2
## # i 1 more variable: median_mld <dbl>
```

```r
vars <- c("gtv", "ptv", "td", "n_fx", "eqd2", "mld")
results_list <- list()

for (var in vars) {
  values <- chart_data[[var]]
  group <- chart_data$Desc_Salv_Consol

  df <- data.frame(value = values, group = group) %>%
```

```r
    filter(!is.na(value), !is.na(group))

  summary_stats <- df %>%
    group_by(group) %>%
    summarise(
      Median = median(value),
      Q1 = quantile(value, 0.25),
      Q3 = quantile(value, 0.75),
      .groups = "drop"
    )

  p_val <- if (length(unique(df$group)) == 2) {
    wilcox.test(value ~ group, data = df)$p.value
  } else {
    kruskal.test(value ~ group, data = df)$p.value
  }

  results_list[[var]] <- data.frame(
    Variable = var,
    Group1 = summary_stats$group[1],
    Median1 = summary_stats$Median[1],
    IQR1 = paste0("(", summary_stats$Q1[1], "-", summary_stats$Q3[1], ")"),
    Group2 = summary_stats$group[2],
    Median2 = summary_stats$Median[2],
    IQR2 = paste0("(", summary_stats$Q1[2], "-", summary_stats$Q3[2], ")"),
    P_value = round(p_val, 3)
  )
}

summary_table <- do.call(rbind, results_list)
summary_table
```

```
##      Variable  Group1  Median1                  IQR1          Group2 Median2
## gtv       gtv Salvage 25.60000             (10-47.2) Consolidate/CMV  26.100
## ptv       ptv Salvage 73.30000            (49.5-140) Consolidate/CMV  94.450
## td         td Salvage 50.00000               (40-50) Consolidate/CMV  43.500
## n_fx     n_fx Salvage  5.00000                 (5-5) Consolidate/CMV   5.000
## eqd2     eqd2 Salvage 83.33333 (60-83.3333333333333) Consolidate/CMV  65.625
## mld       mld Salvage  8.11500          (6.1-10.768) Consolidate/CMV   6.932
##                        IQR2 P_value
## gtv             (18-38.55)   0.561
## ptv           (66.3-153.4)   0.167
## td                 (40-50)   0.283
## n_fx                 (5-5)   0.054
## eqd2 (60-83.3333333333333)   0.150
## mld           (5.282-9.95)   0.372
```

5.6 Total summary table

```r
vars <- c("gtv", "ptv", "td", "n_fx", "eqd2", "mld")
results_list <- list()
```

```r
for (var in vars) {
  values <- chart_data[[var]]
  group <- chart_data$Desc_Salv_Consol

  df <- data.frame(value = values, group = group) %>%
    filter(!is.na(value), !is.na(group))

  summary_stats <- df %>%
    group_by(group) %>%
    summarise(
      Median = median(value),
      Q1 = quantile(value, 0.25),
      Q3 = quantile(value, 0.75),
      .groups = "drop"
    )

  total_summary <- data.frame(
    group = "Total",
    Median = median(df$value),
    Q1 = quantile(df$value, 0.25),
    Q3 = quantile(df$value, 0.75)
  )

  p_val <- if (length(unique(df$group)) == 2) {
    wilcox.test(value ~ group, data = df)$p.value
  } else {
    kruskal.test(value ~ group, data = df)$p.value
  }

  combined <- bind_rows(summary_stats, total_summary) %>%
    mutate(
      Variable = var,
      IQR = paste0("(", round(Q1, 1), "-", round(Q3, 1), ")"),
      P_value = ifelse(group == "Total", NA, round(p_val, 3))
    ) %>%
    select(Variable, Group = group, Median, IQR, P_value)

  results_list[[var]] <- combined
}

final_summary <- bind_rows(results_list)
final_summary
```

```
## # A tibble: 18 x 5
##    Variable Group           Median IQR          P_value
##    <chr>    <chr>            <dbl> <chr>          <dbl>
##  1 gtv      Salvage          25.6  (10-47.2)      0.561
##  2 gtv      Consolidate/CMV  26.1  (18-38.5)      0.561
##  3 gtv      Total            25.6  (12.5-44.5)   NA
##  4 ptv      Salvage          73.3  (49.5-140)     0.167
##  5 ptv      Consolidate/CMV  94.4  (66.3-153.4)   0.167
##  6 ptv      Total            86.8  (56-140)      NA
##  7 td       Salvage          50    (40-50)        0.283
```

```
##  8 td       Consolidate/CMV  43.5  (40-50)        0.283
##  9 td       Total            47.5  (40-50)       NA
## 10 n_fx     Salvage           5    (5-5)          0.054
## 11 n_fx     Consolidate/CMV   5    (5-5)          0.054
## 12 n_fx     Total             5    (5-5)         NA
## 13 eqd2     Salvage          83.3  (60-83.3)      0.15
## 14 eqd2     Consolidate/CMV  65.6  (60-83.3)      0.15
## 15 eqd2     Total            77.2  (60-83.3)     NA
## 16 mld      Salvage           8.12 (6.1-10.8)     0.372
## 17 mld      Consolidate/CMV   6.93 (5.3-10)       0.372
## 18 mld      Total             8    (5.6-10.4)    NA
```

5.7 Correct SBRT end date for one observation

```r
hcc_data <- hcc_data %>%
  mutate(
    sbrt_end = if_else(
      unique_id == 77,
      as.Date("2018-10-19"),
      sbrt_end
    )
  )

hcc_data %>%
  filter(unique_id == 77) %>%
  select(unique_id, date_last_LDT, sbrt_end)
```

```
## # A tibble: 1 x 3
##   unique_id date_last_LDT sbrt_end
##       <dbl> <date>        <date>
## 1        77 2018-08-14    2018-10-19
```

5.8 Additional descriptive tables

```r
table(chart_data$prior_LDT)
```

```
##
##  1  2  3
## 68 18  3
```

```r
fisher.test(table(chart_data$prior_LDT, chart_data$Desc_Salv_Consol))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(chart_data$prior_LDT, chart_data$Desc_Salv_Consol)
## p-value = 0.5629
## alternative hypothesis: two.sided
```

```r
chart_data %>%
  summarise(
    median_prior_LDT = median(prior_LDT, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 1
##   median_prior_LDT
##              <dbl>
## 1                1
```

```r
table(chart_data$last_LDT, chart_data$Desc_Salv_Consol)
```

```
##
##            Salvage Consolidate/CMV
##   DEB-TACE       0               3
##   MWA            3               0
##   RFA            2               0
##   TACE          43              21
##   TAE            3               4
##   Y-90           6               4
```

```r
fisher.test(table(chart_data$last_LDT, chart_data$Desc_Salv_Consol))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(chart_data$last_LDT, chart_data$Desc_Salv_Consol)
## p-value = 0.08487
## alternative hypothesis: two.sided
```

6 Local control analysis

6.1 Build analysis dataset

```r
model_set1 <- chart_data %>%
  mutate(
    time_lc_yr = as.numeric(difftime(lc_censor, sbrt_end, units = "days")) / 365,
    resp = coalesce(resp, ""),
    resp_fail = if_else(str_to_upper(str_trim(resp)) == "CPD", 1, 0)
  )

table(model_set1$resp, model_set1$resp_fail, useNA = "ifany")
```

```
##
##        0  1
##   0    3  0
##   cCR 36  0
##   cPD  0  6
```

```
##   cPR  9  0
##   cSD  5  0
##   pCR 18  0
##   pPR 12  0
```

```r
table(model_set1$resp, useNA = "ifany")
```

```
##
##   0 cCR cPD cPR cSD pCR pPR
##   3  36   6   9   5  18  12
```

```r
table(model_set1$resp_fail, useNA = "ifany")
```

```
##
##  0  1
## 83  6
```

6.2 Truncated 3-year event indicator

```r
model_set1_3yr <- model_set1 %>%
  mutate(
    time_3yr = pmin(time_lc_yr, 3, na.rm = TRUE),
    status_3yr = ifelse(resp_fail == 1 & time_lc_yr <= 3, 1, 0)
  )

table(model_set1_3yr$status_3yr, useNA = "ifany")
```

```
##
##  0  1
## 85  4
```

```r
table(model_set1$resp_fail, model_set1$Desc_Salv_Consol, useNA = "ifany")
```

```
##
##     Salvage Consolidate/CMV
##   0      53              30
##   1       4               2
```

6.3 Follow-up summary

```r
summary(model_set1$time_lc_yr)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02466 0.50411 0.90685 1.50155 1.86301 7.25479
```

```r
max(model_set1$time_lc_yr, na.rm = TRUE)
```

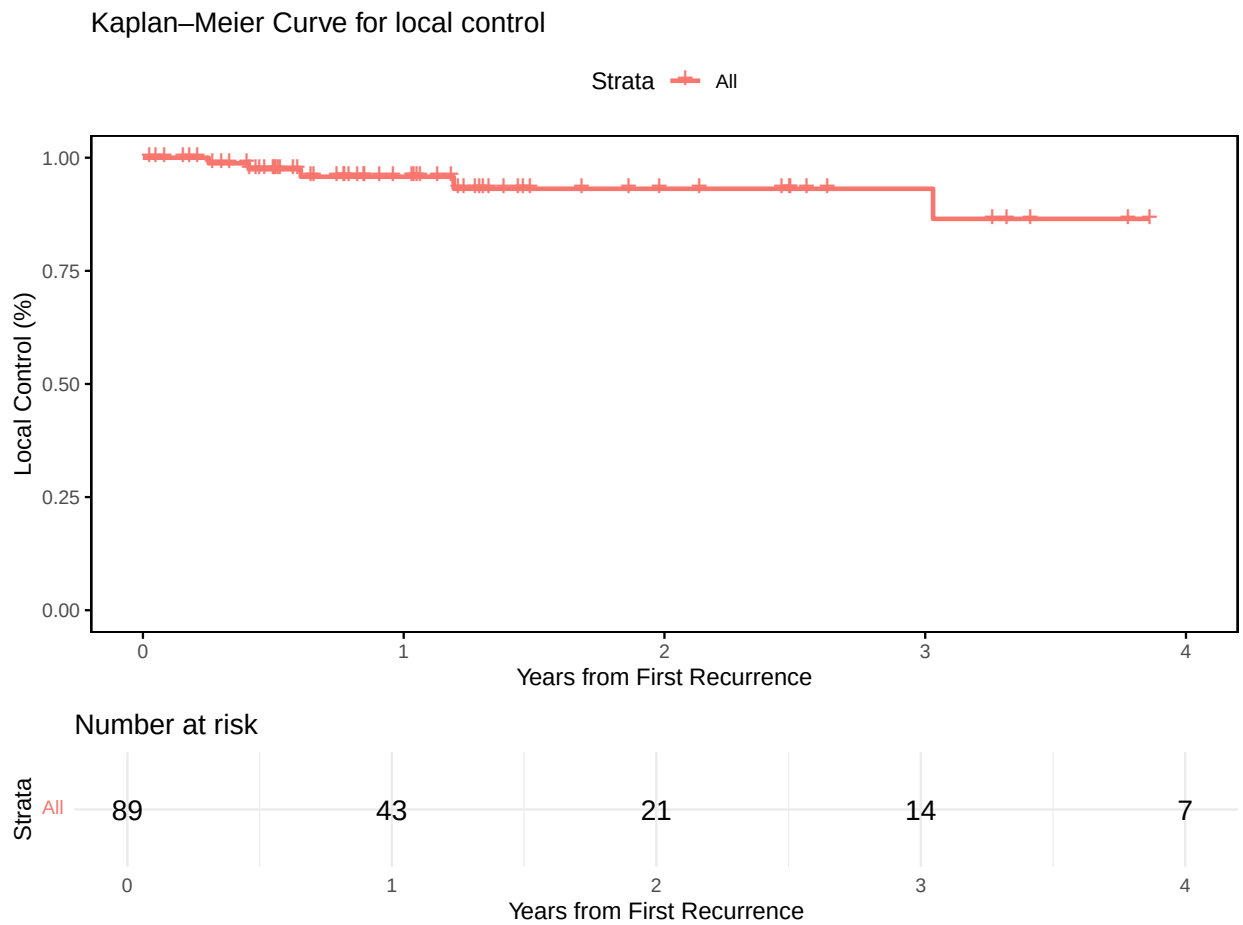
```
## [1] 7.254795
```

```r
model_set1 %>%
  summarise(
    n          = n(),
    min_follow = min(time_lc_yr, na.rm = TRUE),
    q1         = quantile(time_lc_yr, .25, na.rm = TRUE),
    median     = median(time_lc_yr, na.rm = TRUE),
    q3         = quantile(time_lc_yr, .75, na.rm = TRUE),
    max_follow = max(time_lc_yr, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 6
##       n min_follow    q1 median    q3 max_follow
##   <int>      <dbl> <dbl>  <dbl> <dbl>      <dbl>
## 1    89     0.0247 0.504  0.907  1.86       7.25
```

6.4 Overall local control KM

```r
surv_obj <- Surv(time = model_set1$time_lc_yr,
                 event = model_set1$resp_fail)

fit_km <- survfit(surv_obj ~ 1, data = model_set1)

plot_km <- ggsurvplot(
  fit_km,
  conf.int      = FALSE,
  risk.table    = TRUE,
  xlab          = "Years from First Recurrence",
  ylab          = "Local Control (%)",
  title         = "Kaplan-Meier Curve for local control",
  xlim          = c(0, 4),
  break.time.by = 1,
  ggtheme       = theme_minimal()
)

plot_km$plot <- plot_km$plot +
  scale_x_continuous(limits = c(0, 4), breaks = 0:4) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(plot_km)
```




```r
summary(fit_km)
```

```
## Call: survfit(formula = surv_obj ~ 1, data = model_set1)
##
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##  0.252     81       1    0.988  0.0123        0.964            1
##  0.405     75       1    0.974  0.0178        0.940            1
##  0.605     59       1    0.958  0.0240        0.912            1
##  1.192     36       1    0.931  0.0351        0.865            1
##  3.030     14       1    0.865  0.0719        0.735            1
##  4.515      6       1    0.721  0.1446        0.486            1
```

```r
summary(fit_km, times = c(1, 2, 3, 4))
```

```
## Call: survfit(formula = surv_obj ~ 1, data = model_set1)
##
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     43       3    0.958  0.0240        0.912            1
##     2     21       1    0.931  0.0351        0.865            1
##     3     14       0    0.931  0.0351        0.865            1
##     4      7       1    0.865  0.0719        0.735            1
```

```r
median_lc <- summary(fit_km)$table["median"]
median_lc
```

```
## median
##     NA
```

6.5 Local control by treatment group

```r
surv_obj_full <- Surv(time = model_set1$time_lc_yr, event = model_set1$resp_fail)

fit_km_full <- survfit(surv_obj_full ~ 1, data = model_set1, conf.type = "log-log")
summary(fit_km_full, times = c(1, 2, 3, 4))
```

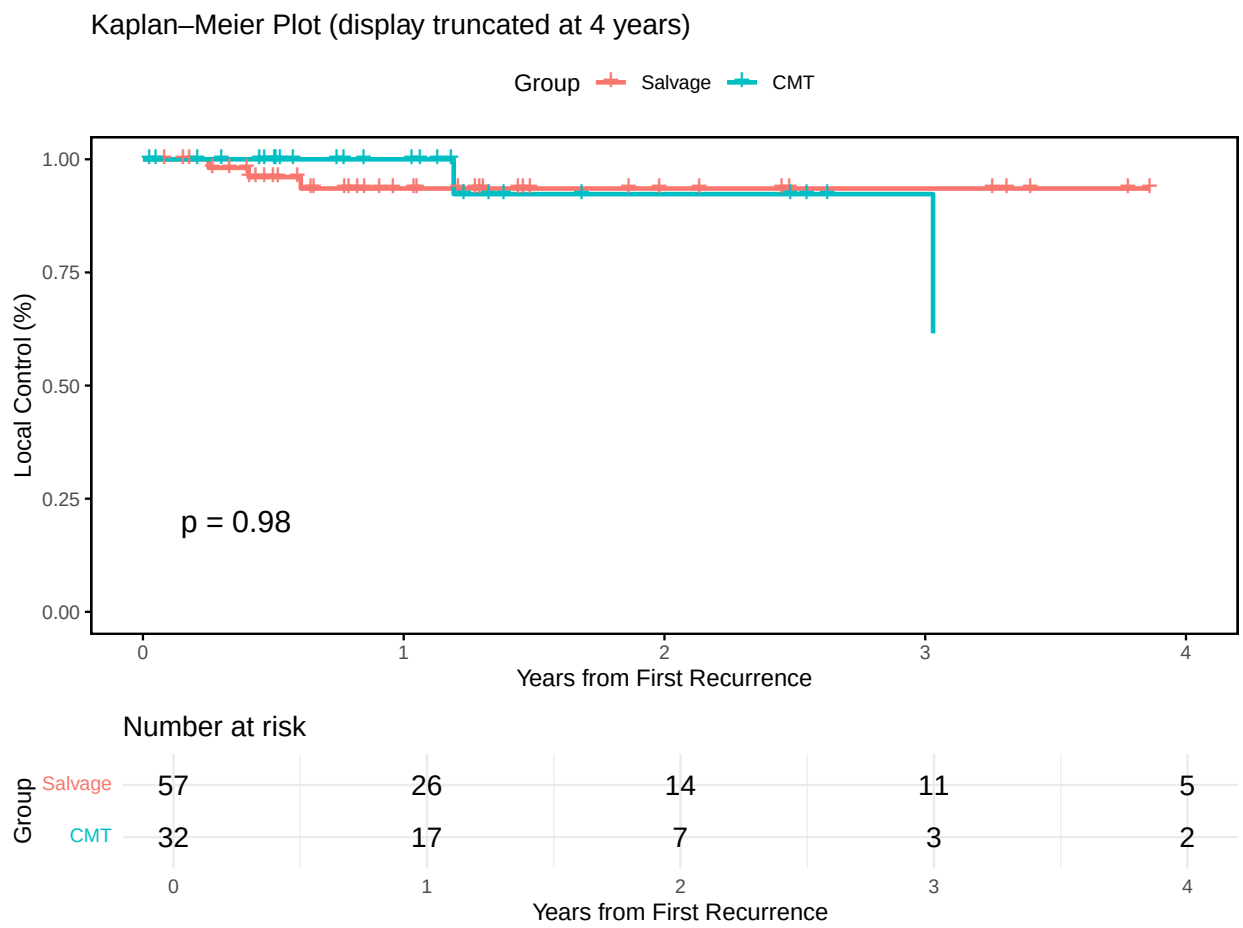
```
## Call: survfit(formula = surv_obj_full ~ 1, data = model_set1, conf.type = "log-log")
##
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     43       3    0.958  0.0240        0.874        0.986
##     2     21       1    0.931  0.0351        0.818        0.975
##     3     14       0    0.931  0.0351        0.818        0.975
##     4      7       1    0.865  0.0719        0.640        0.954
```

```r
fit_stratified <- survfit(Surv(time_lc_yr, resp_fail) ~ Desc_Salv_Consol, data = model_set1)
summary(fit_stratified)
```

```
## Call: survfit(formula = Surv(time_lc_yr, resp_fail) ~ Desc_Salv_Consol,
##     data = model_set1)
##
##                 Desc_Salv_Consol=Salvage
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##  0.252     53       1    0.981  0.0187        0.945            1
##  0.405     48       1    0.961  0.0273        0.909            1
##  0.605     38       1    0.935  0.0364        0.867            1
##  4.515      4       1    0.702  0.2044        0.396            1
##
##                 Desc_Salv_Consol=Consolidate/CMV
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##  1.19     13       1    0.923  0.0739        0.789            1
##  3.03      3       1    0.615  0.2560        0.272            1
```

```r
plot_km <- ggsurvplot(
  fit_stratified,
  conf.int   = FALSE,
  risk.table = TRUE,
  pval       = TRUE,
  xlab       = "Years from First Recurrence",
  ylab       = "Local Control (%)",
  title      = "Kaplan-Meier Plot (display truncated at 4 years)",
  xlim       = c(0, 4),
  break.time.by = 1,
```

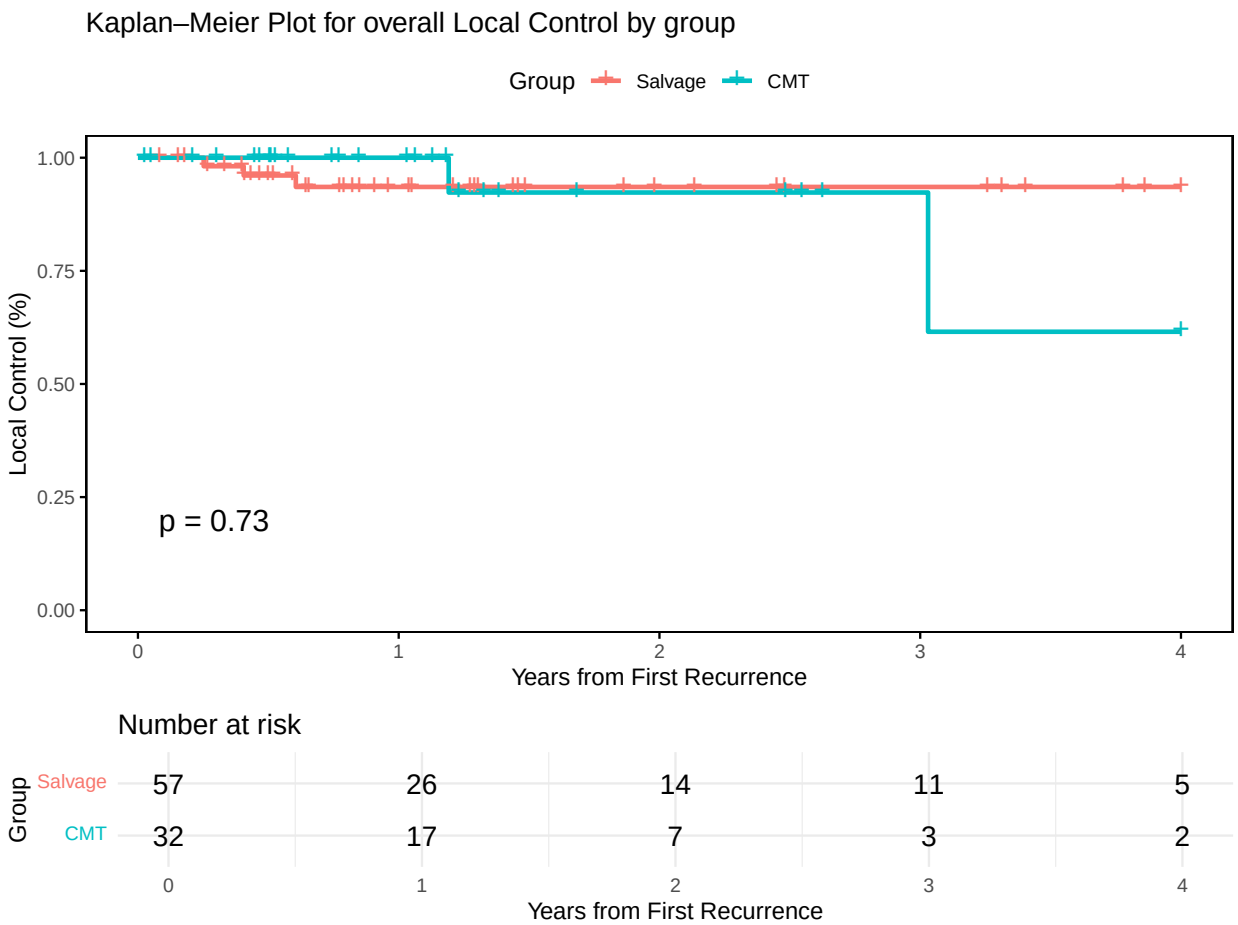
```
  legend.title  = "Group",
  legend.labs   = c("Salvage", "CMT"),
  ggtheme       = theme_minimal()
)

plot_km$plot <- plot_km$plot +
  scale_x_continuous(limits = c(0, 4), breaks = 0:4) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(plot_km)
```



6.6 4-year censored local control by group

```
model_set1$time_lc_yr_4yr  <- pmin(model_set1$time_lc_yr, 4)
model_set1$event_4yr       <- ifelse(model_set1$time_lc_yr > 4, 0, model_set1$resp_fail)

fit_stratified <- survfit(Surv(time_lc_yr_4yr, event_4yr) ~ Desc_Salv_Consol, data = model_set1)
```

```r
plot_km <- ggsurvplot(
  fit_stratified,
  conf.int      = FALSE,
  risk.table    = TRUE,
  pval          = TRUE,
  xlab          = "Years from First Recurrence",
  ylab          = "Local Control (%)",
  title         = "Kaplan-Meier Plot for overall Local Control by group",
  xlim          = c(0, 4),
  break.time.by = 1,
  legend.title  = "Group",
  legend.labs   = c("Salvage", "CMT"),
  ggtheme       = theme_minimal()
)

plot_km$plot <- plot_km$plot +
  scale_x_continuous(limits = c(0, 4), breaks = seq(0, 4, 1)) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(plot_km)
```



6.7 Local control probability table and RMST

```
fit_stratified <- survfit(Surv(time_lc_yr, resp_fail) ~ Desc_Salv_Consol, data = model_set1)

km_summary <- summary(
  fit_stratified,
  times = c(1, 2, 3, 4),
  censored = TRUE,
  extend = TRUE
)

fit_stratified_loglog <- survfit(
  surv_obj_full ~ Desc_Salv_Consol,
  data = model_set1,
  conf.type = "log-log"
)

km_summary_1 <- summary(fit_stratified_loglog, times = c(1, 2, 3, 4))
km_summary_1
```

```
## Call: survfit(formula = surv_obj_full ~ Desc_Salv_Consol, data = model_set1,
```

```
##     conf.type = "log-log")
##
##                 Desc_Salv_Consol=Salvage
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     26       3    0.935  0.0364        0.811        0.979
##     2     14       0    0.935  0.0364        0.811        0.979
##     3     11       0    0.935  0.0364        0.811        0.979
##     4      5       0    0.935  0.0364        0.811        0.979
##
##                 Desc_Salv_Consol=Consolidate/CMV
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     17       0    1.000  0.0000           NA           NA
##     2      7       1    0.923  0.0739        0.566        0.989
##     3      3       0    0.923  0.0739        0.566        0.989
##     4      2       1    0.615  0.2560        0.074        0.913
```

```r
km_df <- data.frame(
  Time_Year = km_summary$time,
  Group     = gsub("Desc_Salv_Consol=", "", km_summary$strata),
  N_Risk    = km_summary$n.risk,
  N_Event   = km_summary$n.event,
  Survival_Percent = round(100 * km_summary$surv, 1),
  CI_Lower_Percent = round(100 * km_summary$lower, 1),
  CI_Upper_Percent = round(100 * km_summary$upper, 1)
)

print(km_df)
```

```
##   Time_Year           Group N_Risk N_Event Survival_Percent CI_Lower_Percent
## 1         1         Salvage     26       3             93.5             86.7
## 2         2         Salvage     14       0             93.5             86.7
## 3         3         Salvage     11       0             93.5             86.7
## 4         4         Salvage      5       0             93.5             86.7
## 5         1 Consolidate/CMV     17       0            100.0            100.0
## 6         2 Consolidate/CMV      7       1             92.3             78.9
## 7         3 Consolidate/CMV      3       0             92.3             78.9
## 8         4 Consolidate/CMV      2       1             61.5             27.2
##   CI_Upper_Percent
## 1              100
## 2              100
## 3              100
## 4              100
## 5              100
## 6              100
## 7              100
## 8              100
```

```r
tau_lc <- 2.5

sum_lc_overall <- summary(fit_km, rmean = tau_lc)$table

overall_lc_rmst <- tibble(
  group     = "Overall",
```

```r
  n         = as.numeric(sum_lc_overall["n"]),
  tau       = tau_lc,
  rmst      = as.numeric(sum_lc_overall["rmean"]),
  se_rmst   = as.numeric(sum_lc_overall["se(rmean)"])
) |>
  mutate(
    lower_95 = rmst - 1.96 * se_rmst,
    upper_95 = rmst + 1.96 * se_rmst
  ) |>
  mutate(across(c(tau, rmst, se_rmst, lower_95, upper_95), ~ round(., 2)))

overall_lc_rmst
```

```
## # A tibble: 1 x 7
##   group       n   tau  rmst se_rmst lower_95 upper_95
##   <chr>   <dbl> <dbl> <dbl>   <dbl>    <dbl>    <dbl>
## 1 Overall    NA   2.5  2.38    0.06     2.26     2.49
```

```r
rmst_lc_tbl <- summary(fit_stratified, rmean = tau_lc)$table |>
  as.data.frame() |>
  rownames_to_column("group") |>
  transmute(
    group,
    n        = records,
    events   = events,
    tau      = tau_lc,
    rmst     = rmean,
    se_rmst  = `se(rmean)`,
    lower_95 = rmean - 1.96 * `se(rmean)`,
    upper_95 = rmean + 1.96 * `se(rmean)`
  ) |>
  mutate(across(c(tau, rmst, se_rmst, lower_95, upper_95), ~ round(., 2)))

rmst_lc_tbl
```

```
##                              group  n events tau rmst se_rmst lower_95 upper_95
## 1         Desc_Salv_Consol=Salvage 57      4 2.5 2.37    0.07     2.22     2.51
## 2 Desc_Salv_Consol=Consolidate/CMV 32      2 2.5 2.40    0.10     2.21     2.59
```

7 Overall survival analysis

7.1 Build OS dataset

```r
model_overall <- studyid_wise %>%
  mutate(
    time_os = as.numeric(os_censor - date_LTD1) / 365,
    flg_over = ifelse(os_censor_reason == "died", 1, 0)
  )
table(model_overall$flg_over, useNA = "ifany")
```

```
##
##  0  1
## 28 49
```

```r
nrow(model_overall)
```

```
## [1] 77
```

```r
model_overall <- studyid_wise %>%
  mutate(
    time_os   = as.numeric(os_censor - date_LTD1) / 365,
    status_os = ifelse(os_censor_reason == "died", 1, 0)
  )

table(model_overall$status_os, useNA = "ifany")
```

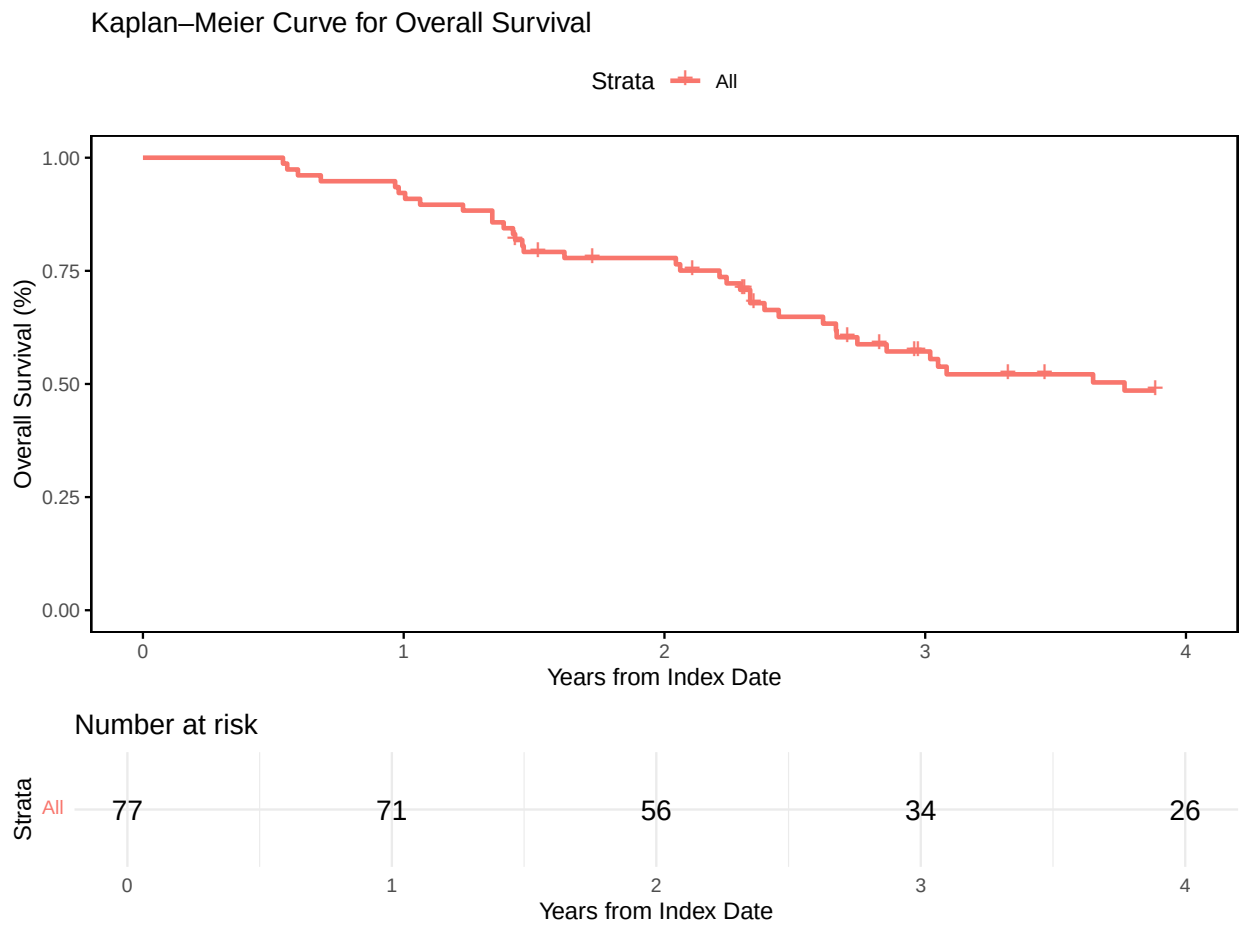
```
##
##  0  1
## 28 49
```

```r
summary(model_overall$time_os)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.537   1.723   2.740   3.617   5.016  12.460
```

7.2 Overall OS KM, median, RMST

```r
fit_os <- survfit(Surv(time_os, status_os) ~ 1, data = model_overall)

plot_os <- ggsurvplot(
  fit_os,
  conf.int      = FALSE,
  risk.table    = TRUE,
  xlab          = "Years from Index Date",
  ylab          = "Overall Survival (%)",
  title         = "Kaplan-Meier Curve for Overall Survival",
  xlim          = c(0, 4),
  break.time.by = 1,
  ggtheme       = theme_minimal()
)

plot_os$plot <- plot_os$plot +
  scale_x_continuous(limits = c(0, 4), breaks = 0:4) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(plot_os)
```




```
times <- c(1, 2, 3, 4)
km_at_t <- summary(fit_os, times = times)

prob_tbl <- data.frame(
  time      = km_at_t$time,
  surv_prob = km_at_t$surv,
  lower_95  = km_at_t$lower,
  upper_95  = km_at_t$upper
)

prob_tbl
```

```
##   time surv_prob  lower_95  upper_95
## 1    1 0.9220779 0.8641092 0.9839355
## 2    2 0.7783687 0.6907135 0.8771479
## 3    3 0.5718328 0.4672577 0.6998125
## 4    4 0.4854199 0.3787776 0.6220867
```

```
med_vec <- summary(fit_os)$table[c("median", "0.95LCL", "0.95UCL")]

median_tbl <- data.frame(
  median_surv = med_vec["median"],
  lower_95    = med_vec["0.95LCL"],
```

```
  upper_95    = med_vec["0.95UCL"]
)

median_tbl
```

```
##        median_surv lower_95 upper_95
## median    3.764384 2.660274      5.4
```

```
tau_os <- 7.0

sum_os_7 <- summary(fit_os, rmean = tau_os)$table
rmst_overall_os_7 <- data.frame(
  tau        = tau_os,
  rmst_total = as.numeric(sum_os_7["rmean"]),
  se_rmst    = as.numeric(sum_os_7["se(rmean)"])
)

rmst_overall_os_7$lower_95 <- with(rmst_overall_os_7, rmst_total - 1.96 * se_rmst)
rmst_overall_os_7$upper_95 <- with(rmst_overall_os_7, rmst_total + 1.96 * se_rmst)
rmst_overall_os_7[] <- lapply(rmst_overall_os_7, \(x) if(is.numeric(x)) round(x, 2) else x)

rmst_overall_os_7
```

```
##   tau rmst_total se_rmst lower_95 upper_95
## 1   7       4.07    0.28     3.52     4.63
```

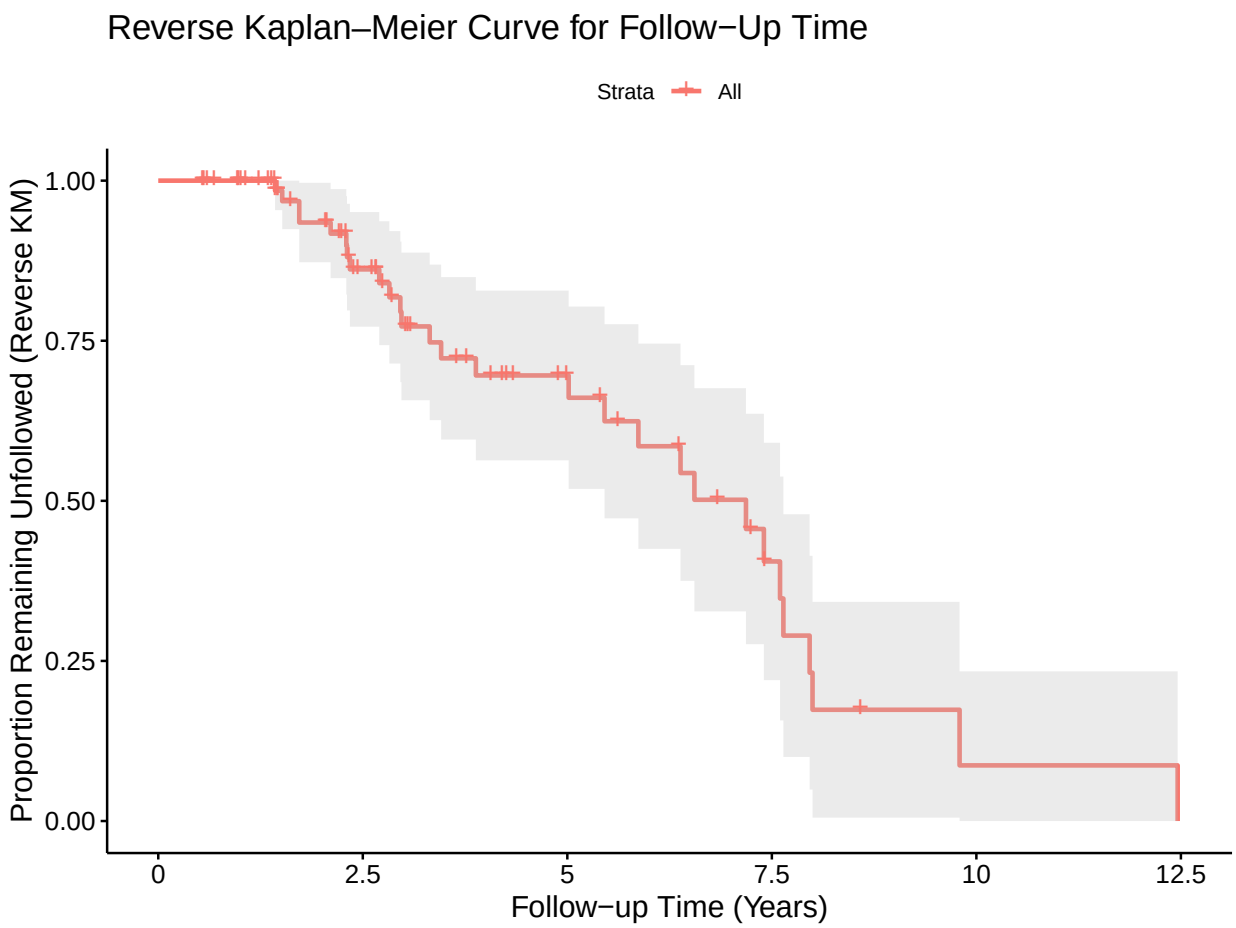
7.3 Reverse KM follow-up

```
model_overall <- model_overall %>%
  mutate(followup_surv = Surv(time_os, status_os == 0))

rev_km_fit_total <- survfit(followup_surv ~ 1, data = model_overall, conf.type = "plain")
rev_km_total <- summary(rev_km_fit_total)$table[c("median", "0.95LCL", "0.95UCL")]

followup_total_tbl <- data.frame(
  Group            = "Total",
  Median_Followup  = rev_km_total["median"],
  Lower_95_CI      = rev_km_total["0.95LCL"],
  Upper_95_CI      = rev_km_total["0.95UCL"]
)

rev_km_fit_by_group <- survfit(followup_surv ~ Desc_Salv_Consol, data = model_overall, conf.type = "pla
group_summaries <- summary(rev_km_fit_by_group)$table

followup_group_tbl <- data.frame(
  Group            = rownames(group_summaries),
  Median_Followup  = group_summaries[, "median"],
  Lower_95_CI      = group_summaries[, "0.95LCL"],
  Upper_95_CI      = group_summaries[, "0.95UCL"]
)
```

```r
full_followup_tbl <- bind_rows(followup_total_tbl, followup_group_tbl)
full_followup_tbl
```

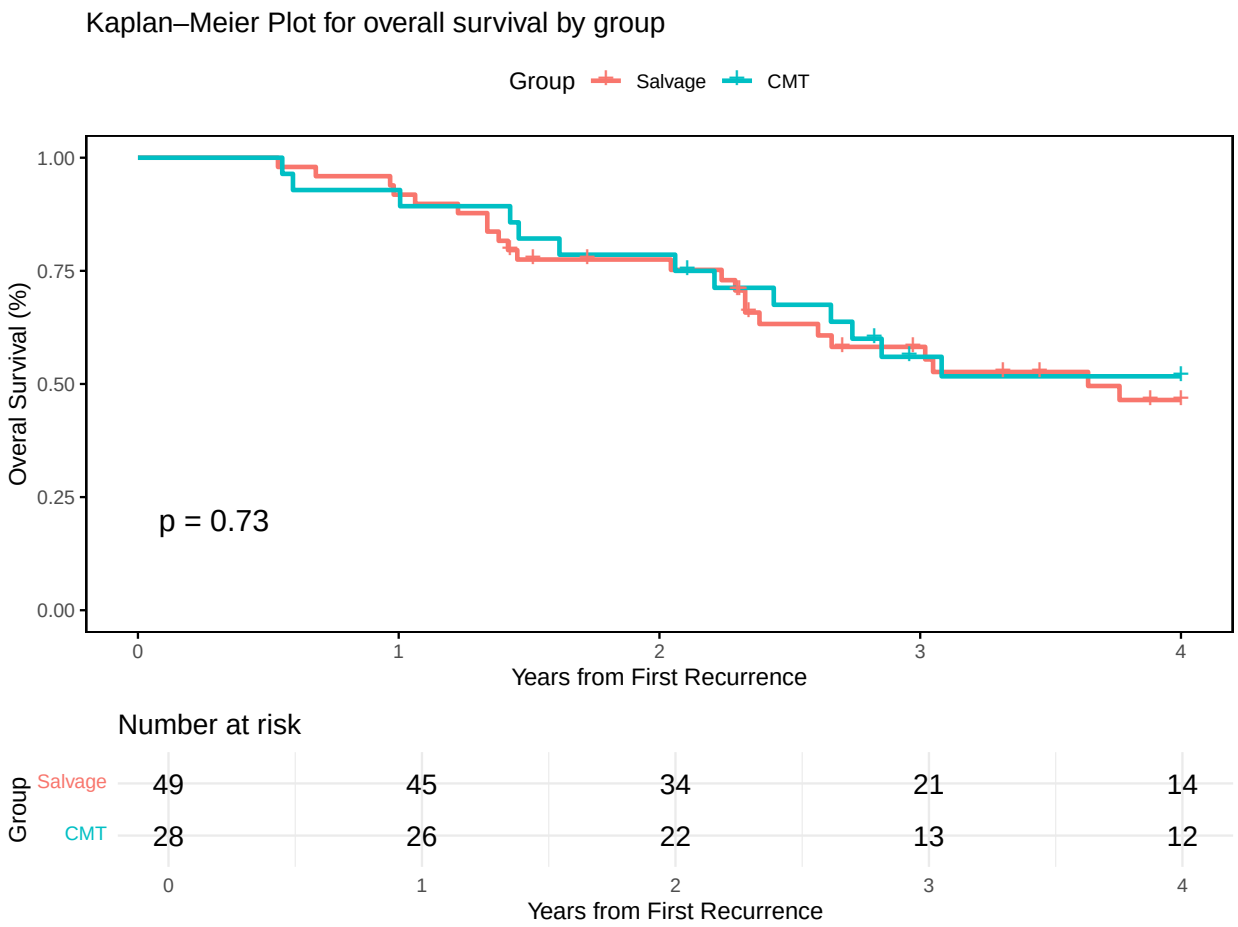
```
##                                                                Group
## median                                                         Total
## Desc_Salv_Consol=Salvage                  Desc_Salv_Consol=Salvage
## Desc_Salv_Consol=Consolidate/CMV Desc_Salv_Consol=Consolidate/CMV
##                                  Median_Followup Lower_95_CI Upper_95_CI
## median                                  7.183562    5.454795    7.641096
## Desc_Salv_Consol=Salvage                5.868493    3.457534    9.794521
## Desc_Salv_Consol=Consolidate/CMV        7.402740    6.383562    7.641096
```

```r
ggsurvplot(
  rev_km_fit_total,
  conf.int = TRUE,
  xlab = "Follow-up Time (Years)",
  ylab = "Proportion Remaining Unfollowed (Reverse KM)",
  title = "Reverse Kaplan-Meier Curve for Follow-Up Time"
)
```



7.4 Overall survival by treatment group

```
fit_os <- survfit(Surv(time_os, status_os) ~ 1, data = model_overall)
model_overall$time_os_4yr <- pmin(model_overall$time_os, 4)
model_overall$event_os_4yr <- ifelse(model_overall$time_os > 4, 0, model_overall$status_os)

fit_stratified_os <- survfit(Surv(time_os_4yr, event_os_4yr) ~ Desc_Salv_Consol, data = model_overall)

plot_km <- ggsurvplot(
  fit_stratified_os,
  conf.int      = FALSE,
  risk.table    = TRUE,
  pval          = TRUE,
  xlab          = "Years from First Recurrence",
  ylab          = "Overal Survival (%)",
  title         = "Kaplan-Meier Plot for overall survival by group",
  xlim          = c(0, 4),
  break.time.by = 1,
  legend.title  = "Group",
  legend.labs   = c("Salvage", "CMT"),
  ggtheme       = theme_minimal()
)

plot_km$plot <- plot_km$plot +
  scale_x_continuous(limits = c(0, 4), breaks = seq(0, 4, 1)) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(plot_km)
```



8 Subgroup analysis

8.1 Treatment interval subgroup for local control

```
model_subgroup <- chart_data %>%
  mutate(
    months_treat_time = as.numeric(difftime(sbrt_end, date_last_LDT, units = "days")) / 30.44,
    subgroup_flg = case_when(
      months_treat_time <  6 ~ 1,
      months_treat_time >= 6 ~ 2,
      TRUE                   ~ NA_real_
    ),
    time_lc_yr = as.numeric(difftime(lc_censor, sbrt_end, units = "days")) / 365,
    resp_fail  = if_else(str_to_upper(str_trim(coalesce(resp, ""))) == "CPD", 1, 0),
    treatment_group = factor(
      subgroup_flg,
      levels = c(1, 2),
      labels = c("<6 months treatment", " 6 months treatment")
    )
  )
```

```r
surv_obj <- Surv(time = model_subgroup$time_lc_yr,
                 event = model_subgroup$resp_fail)

fit_km_sg <- survfit(surv_obj ~ treatment_group, data = model_subgroup)

summary(fit_km_sg, times = c(1, 2, 3, 4))
```

```
## Call: survfit(formula = surv_obj ~ treatment_group, data = model_subgroup)
##
##                 treatment_group=<6 months treatment
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     29       2    0.959  0.0287        0.904            1
##     2     16       1    0.922  0.0455        0.837            1
##     3      9       0    0.922  0.0455        0.837            1
##     4      5       1    0.820  0.1047        0.638            1
##
##                 treatment_group= 6 months treatment
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     14       1    0.955  0.0444        0.871            1
##     2      5       0    0.955  0.0444        0.871            1
##     3      5       0    0.955  0.0444        0.871            1
##     4      2       0    0.955  0.0444        0.871            1
```

```r
summary(fit_km_sg, times = 1:4, conf.type = "plain")
```

```
## Call: survfit(formula = surv_obj ~ treatment_group, data = model_subgroup)
##
##                 treatment_group=<6 months treatment
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     29       2    0.959  0.0287        0.904            1
##     2     16       1    0.922  0.0455        0.837            1
##     3      9       0    0.922  0.0455        0.837            1
##     4      5       1    0.820  0.1047        0.638            1
##
##                 treatment_group= 6 months treatment
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     14       1    0.955  0.0444        0.871            1
##     2      5       0    0.955  0.0444        0.871            1
##     3      5       0    0.955  0.0444        0.871            1
##     4      2       0    0.955  0.0444        0.871            1
```

```r
summary(fit_km_sg, conf.type = "plain")$table
```

```
##                                    records n.max n.start events    rmean
## treatment_group=<6 months treatment      62    62      62      4 6.319877
## treatment_group= 6 months treatment      27    27      27      2 4.328269
##                                    se(rmean)   median 0.95LCL 0.95UCL
## treatment_group=<6 months treatment 0.4878889       NA      NA      NA
## treatment_group= 6 months treatment 0.1825047 4.515068      NA      NA
```

```r
max_by_arm <- with(model_subgroup,
                   tapply(time_lc_yr, treatment_group, max, na.rm = TRUE))
max_by_arm
```

```
## <6 months treatment  6 months treatment
##            7.254795            4.515068
```

```r
tau <- 4.5
tbl <- summary(fit_km_sg, rmean = tau)$table

rmst_tbl <- tbl |>
  as.data.frame(check.names = FALSE) |>
  tibble::rownames_to_column("group") |>
  dplyr::transmute(
    group     = sub("^.*=", "", group),
    tau       = tau,
    rmst      = rmean,
    se_rmst   = `se(rmean)`,
    lower_95  = rmst - 1.96 * se_rmst,
    upper_95  = rmst + 1.96 * se_rmst
  )

rmst_tbl
```

```
##                 group tau     rmst   se_rmst lower_95 upper_95
## 1 <6 months treatment 4.5 4.061736 0.2112139 3.647757 4.475715
## 2  6 months treatment 4.5 4.313885 0.1818355 3.957488 4.670283
```

8.2 Function to pick tau before heavy censoring

```r
cutoff_tau_before_heavy_censoring <- function(fit, by = 0.25, risk_threshold = 0.10) {
  s_all <- summary(fit)
  t_max <- min(tapply(s_all$time, s_all$strata, max, na.rm = TRUE), na.rm = TRUE)
  times <- seq(0, t_max, by = by)

  s <- summary(fit, times = times)

  df <- tibble(
    time     = s$time,
    strata   = s$strata,
    n_risk   = s$n.risk,
    n_event  = s$n.event,
    n_censor = s$n.censor
  )

  n0 <- df %>%
    group_by(strata) %>%
    summarise(N0 = max(n_risk, na.rm = TRUE), .groups = "drop")

  df2 <- df %>%
```

```r
    inner_join(n0, by = "strata") %>%
    mutate(frac_at_risk = n_risk / N0)

  limiting <- df2 %>%
    group_by(time) %>%
    summarise(min_frac_at_risk = min(frac_at_risk, na.rm = TRUE), .groups = "drop")

  tau_suggested <- limiting %>%
    filter(min_frac_at_risk >= risk_threshold) %>%
    summarise(tau = max(time, na.rm = TRUE)) %>%
    pull(tau)

  list(
    tau_suggested       = tau_suggested,
    limiting_support    = limiting,
    per_time_support    = df2,
    t_max_shortest      = t_max,
    risk_threshold_used = risk_threshold,
    grid_by             = by
  )
}

out_sg <- cutoff_tau_before_heavy_censoring(fit_km_sg, by = 0.25, risk_threshold = 0.10)
out_sg$tau_suggested
```

```
## [1] 3
```

```r
head(out_sg$limiting_support, 10)
```

```
## # A tibble: 10 x 2
##     time min_frac_at_risk
##    <dbl>            <dbl>
##  1  0               1
##  2  0.25            0.889
##  3  0.5             0.704
##  4  0.75            0.593
##  5  1               0.468
##  6  1.25            0.355
##  7  1.5             0.222
##  8  1.75            0.222
##  9  2               0.185
## 10  2.25            0.185
```

```r
out_sg_20 <- cutoff_tau_before_heavy_censoring(fit_km_sg, by = 0.25, risk_threshold = 0.20)
out_sg_20$tau_suggested
```

```
## [1] 1.75
```

```r
tbl1 <- summary(fit_km_sg, rmean = out_sg$tau_suggested)$table

rmst_tbl <- tbl1 |>
```

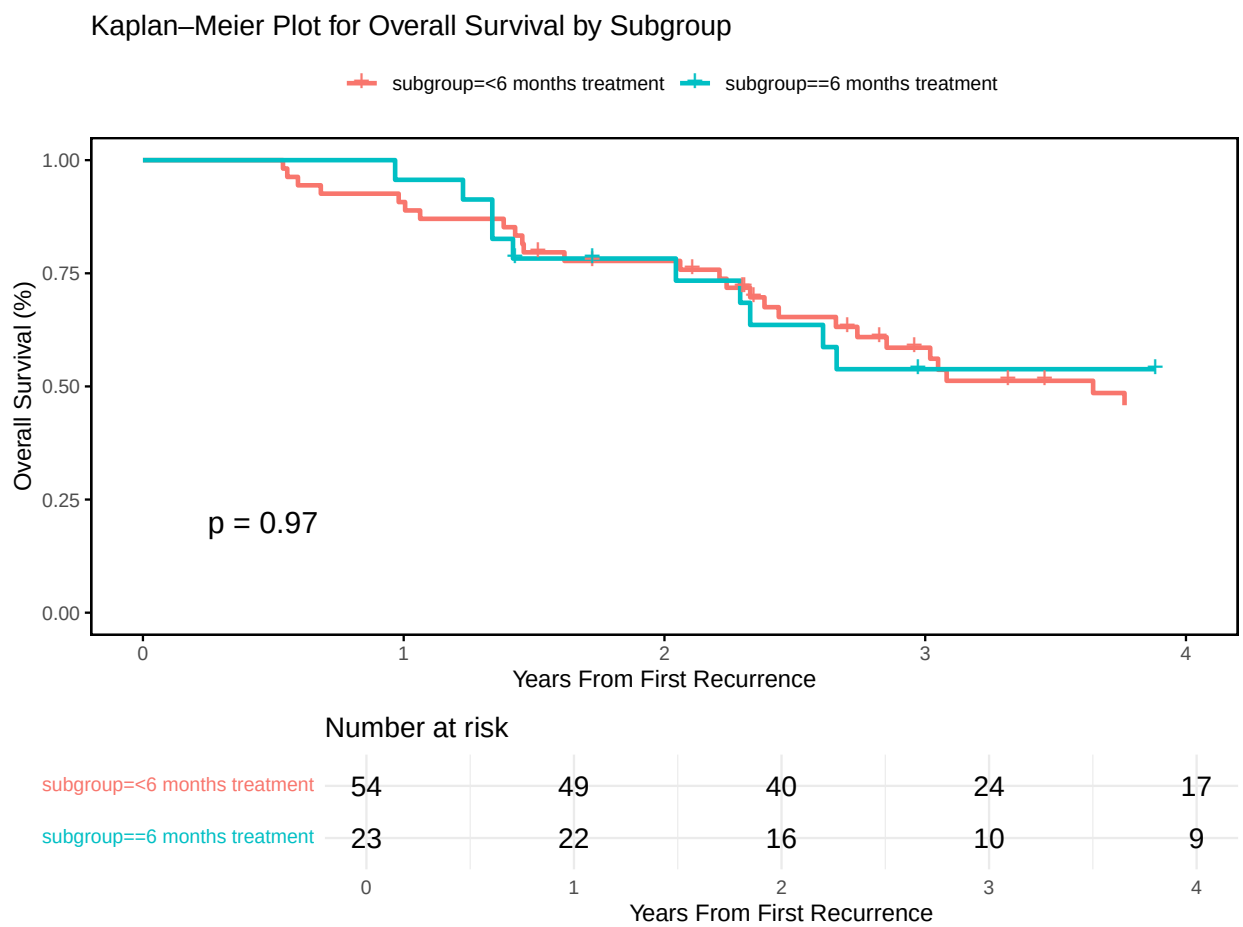
```r
  as.data.frame(check.names = FALSE) |>
  tibble::rownames_to_column("group") |>
  dplyr::transmute(
    group     = sub("^.*=", "", group),
    tau       = tau,
    rmst      = rmean,
    se_rmst   = `se(rmean)`,
    lower_95  = rmst - 1.96 * se_rmst,
    upper_95  = rmst + 1.96 * se_rmst
  )

rmst_tbl
```

```
##                 group tau     rmst    se_rmst lower_95 upper_95
## 1 <6 months treatment 4.5 2.829078 0.09615427 2.640616 3.017541
## 2  6 months treatment 4.5 2.882067 0.11522129 2.656234 3.107901
```

8.3 Overall survival by subgroup

```r
model_surv_sub <- studyid_wise %>%
  mutate(
    time_os = as.numeric(os_censor - date_LTD1) / 365,
    status_os = if_else(os_censor_reason == "died", 1, 0),
    months_treat_time = as.numeric(difftime(sbrt_end, date_last_LDT, units = "days")) / 30.44,
    subgroup_flg = case_when(
      months_treat_time < 6  ~ 1,
      months_treat_time >= 6 ~ 2,
      TRUE ~ NA_real_
    ),
    subgroup = factor(subgroup_flg,
                      levels = c(1, 2),
                      labels = c("<6 months treatment", " 6 months treatment"))
  )

model_surv_sub <- filter(model_surv_sub, !is.na(subgroup))

fit_os_subgroup <- survfit(Surv(time_os, status_os) ~ subgroup, data = model_surv_sub)

plot_os_subgroup <- ggsurvplot(
  fit_os_subgroup,
  conf.int      = FALSE,
  risk.table    = TRUE,
  pval          = TRUE,
  xlab          = "Years From First Recurrence",
  ylab          = "Overall Survival (%)",
  title         = "Kaplan-Meier Plot for Overall Survival by Subgroup",
  legend.title  = "",
  xlim          = c(0, 4),
  break.time.by = 1,
  ggtheme       = theme_minimal()
)
```

```r
plot_os_subgroup$plot <- plot_os_subgroup$plot +
  scale_x_continuous(limits = c(0, 4), breaks = 0:4) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(plot_os_subgroup)
```




```r
log_rank <- survdiff(Surv(time_os, status_os) ~ subgroup, data = model_surv_sub)
pval <- 1 - pchisq(log_rank$chisq, df = length(log_rank$n) - 1)
cat("Log-rank p-value:", signif(pval, 4), "\n")
```

```
## Log-rank p-value: 0.9682
```

8.4 Subgroup survival probabilities and RMST

```r
times_vec <- 1:4
```

```r
fit_overall <- survfit(Surv(time_os, status_os) ~ 1, data = model_surv_sub)
sp_all <- summary(fit_overall, times = times_vec)

overall_tbl <- tibble(
  group     = "All patients",
  time_year = sp_all$time,
  surv_prob = sp_all$surv,
  lower_95  = sp_all$lower,
  upper_95  = sp_all$upper
)

get_probs <- function(dat, label, times = times_vec, conf.type = "log-log") {
  sf  <- survfit(Surv(time_os, status_os) ~ 1, data = dat, conf.type = conf.type)
  ss  <- summary(sf, times = times)
  tibble(
    group     = label,
    time_year = ss$time,
    surv_prob = ss$surv,
    lower_95  = ss$lower,
    upper_95  = ss$upper
  )
}

by_group_tbl <- bind_rows(
  get_probs(filter(model_surv_sub, subgroup == "<6 months treatment"), "<6 months treatment"),
  get_probs(filter(model_surv_sub, subgroup == " 6 months treatment"), " 6 months treatment")
)

surv_probs_tbl <- bind_rows(overall_tbl, by_group_tbl) %>%
  mutate(across(c(surv_prob, lower_95, upper_95), ~ round(.x, 3)))

surv_probs_tbl
```

```
## # A tibble: 12 x 5
##    group               time_year surv_prob lower_95 upper_95
##    <chr>                   <int>     <dbl>    <dbl>    <dbl>
##  1 All patients                1     0.922    0.864    0.984
##  2 All patients                2     0.778    0.691    0.877
##  3 All patients                3     0.572    0.467    0.7
##  4 All patients                4     0.485    0.379    0.622
##  5 <6 months treatment         1     0.907    0.792    0.96
##  6 <6 months treatment         2     0.777    0.641    0.867
##  7 <6 months treatment         3     0.586    0.436    0.708
##  8 <6 months treatment         4     0.458    0.309    0.596
##  9  6 months treatment         1     0.957    0.729    0.994
## 10  6 months treatment         2     0.783    0.554    0.903
## 11  6 months treatment         3     0.538    0.31     0.72
## 12  6 months treatment         4     0.538    0.31     0.72
```

```r
cutoff_tau_before_heavy_censoring <- function(fit, by = 0.25, risk_threshold = 0.10) {
  s_all <- summary(fit)
  t_max <- min(tapply(s_all$time, s_all$strata, max, na.rm = TRUE), na.rm = TRUE)
  times <- seq(0, t_max, by = by)
```

```r
  s <- summary(fit, times = times)

  df <- tibble(
    time     = s$time,
    strata   = s$strata,
    n_risk   = s$n.risk
  )

  n0 <- df %>%
    group_by(strata) %>%
    summarise(N0 = max(n_risk, na.rm = TRUE), .groups = "drop")

  df2 <- df %>%
    inner_join(n0, by = "strata") %>%
    mutate(frac_at_risk = n_risk / N0)

  limiting <- df2 %>%
    group_by(time) %>%
    summarise(min_frac_at_risk = min(frac_at_risk, na.rm = TRUE), .groups = "drop")

  tau_suggested <- limiting %>%
    filter(min_frac_at_risk >= risk_threshold) %>%
    summarise(tau = max(time, na.rm = TRUE)) %>%
    pull(tau)

  list(
    tau_suggested       = tau_suggested,
    limiting_support    = limiting,
    per_time_support    = df2,
    t_max_shortest      = t_max,
    risk_threshold_used = risk_threshold,
    grid_by             = by
  )
}

out_tau <- cutoff_tau_before_heavy_censoring(fit_os_subgroup, by = 0.25, risk_threshold = 0.10)
tau_os_sub <- out_tau$tau_suggested
tau_os_sub
```

```
## [1] 6.5
```

```r
median_tbl <- summary(fit_os_subgroup)$table[, "median", drop = FALSE] %>%
  as.data.frame() %>%
  rownames_to_column("group") %>%
  rename(median_time = median)

rmst_tbl <- summary(fit_os_subgroup, rmean = tau_os_sub)$table %>%
  as.data.frame() %>%
  rownames_to_column("group") %>%
  transmute(
    group,
    n        = records,
    events   = events,
```

```r
    tau      = tau_os_sub,
    rmst     = rmean,
    se_rmst  = `se(rmean)`,
    lower_95 = rmean - 1.96 * se_rmst,
    upper_95 = rmean + 1.96 * se_rmst
  ) %>%
  mutate(across(c(tau, rmst, se_rmst, lower_95, upper_95), ~ round(., 2)))

final_tbl <- median_tbl %>%
  left_join(rmst_tbl, by = "group")

final_tbl
```

```
##                          group median_time  n events tau rmst se_rmst lower_95
## 1 subgroup=<6 months treatment    3.643836 54     35 6.5 3.87    0.31     3.26
## 2 subgroup= 6 months treatment    4.063014 23     14 6.5 4.02    0.47     3.10
##   upper_95
## 1     4.48
## 2     4.94
```

9 Cox regression analyses

```r
model_set1 <- model_set1 %>%
  mutate(
    hcc_size_gt = if_else(hcc_size_sbrt > 3, 1L, 0L)
  )

cox_hccsize <- coxph(
  Surv(time_lc_yr, resp_fail) ~ hcc_size_gt,
  data = model_set1
)
summary(cox_hccsize)
```

```
## Call:
## coxph(formula = Surv(time_lc_yr, resp_fail) ~ hcc_size_gt, data = model_set1)
##
##   n= 89, number of events= 6
##
##              coef exp(coef) se(coef)     z Pr(>|z|)
## hcc_size_gt 1.041     2.833    1.099 0.947    0.344
##
##             exp(coef) exp(-coef) lower .95 upper .95
## hcc_size_gt     2.833      0.353    0.3284     24.44
##
## Concordance= 0.587  (se = 0.098 )
## Likelihood ratio test= 1.1  on 1 df,   p=0.3
## Wald test            = 0.9  on 1 df,   p=0.3
## Score (logrank) test = 0.98  on 1 df,   p=0.3
```

```r
cox_trt_os <- coxph(
  Surv(time_os, status_os) ~ Desc_Salv_Consol,
  data = model_overall
)
summary(cox_trt_os)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol,
##     data = model_overall)
##
##   n= 77, number of events= 49
##
##                                     coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV -0.1603    0.8519   0.2997 -0.535    0.593
##
##                                  exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    0.8519      1.174    0.4735     1.533
##
## Concordance= 0.516  (se = 0.039 )
## Likelihood ratio test= 0.29  on 1 df,   p=0.6
## Wald test            = 0.29  on 1 df,   p=0.6
## Score (logrank) test = 0.29  on 1 df,   p=0.6
```

```r
model_overall <- model_overall %>%
  mutate(age_gt60 = if_else(age_first_SBRT > 60, 1L, 0L))

cox_age_gt60 <- coxph(
  Surv(time_os, status_os) ~ age_gt60,
  data = model_overall
)
summary(cox_age_gt60)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ age_gt60, data = model_overall)
##
##   n= 77, number of events= 49
##
##             coef exp(coef) se(coef)     z Pr(>|z|)
## age_gt60 -0.1554    0.8561   0.3170 -0.49    0.624
##
##          exp(coef) exp(-coef) lower .95 upper .95
## age_gt60    0.8561      1.168    0.4599     1.594
##
## Concordance= 0.508  (se = 0.036 )
## Likelihood ratio test= 0.23  on 1 df,   p=0.6
## Wald test            = 0.24  on 1 df,   p=0.6
## Score (logrank) test = 0.24  on 1 df,   p=0.6
```

```r
model_overall <- model_overall %>%
  mutate(
    Milan_criteria = relevel(factor(Milan_criteria), ref = "No")
  )
```

```r
cox_milan <- coxph(
  Surv(time_os, status_os) ~ Milan_criteria,
  data = model_overall
)
summary(cox_milan)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Milan_criteria, data = model_overall)
##
##   n= 77, number of events= 49
##
##                      coef exp(coef) se(coef)      z Pr(>|z|)
## Milan_criteriaYes -0.2794    0.7562   0.3008 -0.929    0.353
##
##                   exp(coef) exp(-coef) lower .95 upper .95
## Milan_criteriaYes    0.7562      1.322    0.4193     1.364
##
## Concordance= 0.542  (se = 0.037 )
## Likelihood ratio test= 0.84  on 1 df,   p=0.4
## Wald test            = 0.86  on 1 df,   p=0.4
## Score (logrank) test = 0.87  on 1 df,   p=0.4
```

```r
cox_listed <- coxph(
  Surv(time_os, status_os) ~ listed,
  data = model_overall
)
summary(cox_listed)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ listed, data = model_overall)
##
##   n= 77, number of events= 49
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## listedYes -0.8352    0.4338   0.3141 -2.659  0.00783 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##           exp(coef) exp(-coef) lower .95 upper .95
## listedYes    0.4338      2.305    0.2344    0.8028
##
## Concordance= 0.585  (se = 0.039 )
## Likelihood ratio test= 7.56  on 1 df,   p=0.006
## Wald test            = 7.07  on 1 df,   p=0.008
## Score (logrank) test = 7.44  on 1 df,   p=0.006
```

```r
cox_transplant <- coxph(
  Surv(time_os, status_os) ~ transplant,
  data = model_overall
)
summary(cox_transplant)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ transplant, data = model_overall)
##
##   n= 77, number of events= 49
##
##                  coef exp(coef) se(coef)      z Pr(>|z|)
## transplantYes -1.5838    0.2052   0.4033 -3.927 8.59e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## transplantYes    0.2052      4.874   0.09309    0.4523
##
## Concordance= 0.635  (se = 0.031 )
## Likelihood ratio test= 20.18  on 1 df,   p=7e-06
## Wald test            = 15.42  on 1 df,   p=9e-05
## Score (logrank) test = 18.1  on 1 df,   p=2e-05
```

```r
cox_pvtt <- coxph(
  Surv(time_os, status_os) ~ pvtt,
  data = model_overall
)
summary(cox_pvtt)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ pvtt, data = model_overall)
##
##   n= 77, number of events= 49
##
##           coef exp(coef) se(coef)     z Pr(>|z|)
## pvttYes 0.5921    1.8077   0.5265 1.124    0.261
##
##         exp(coef) exp(-coef) lower .95 upper .95
## pvttYes     1.808     0.5532    0.6441     5.074
##
## Concordance= 0.518  (se = 0.021 )
## Likelihood ratio test= 1.08  on 1 df,   p=0.3
## Wald test            = 1.26  on 1 df,   p=0.3
## Score (logrank) test = 1.3  on 1 df,   p=0.3
```

```r
model_overall <- model_overall %>%
  mutate(
    afp_prior_sbrt_cat = if_else(afp_prior_sbrt >= 10, 1L, 0L)
  )

table(model_overall$afp_prior_sbrt_cat, useNA = "ifany")
```

```
##
##  0  1
## 47 30
```

```r
cox_afp <- coxph(
  Surv(time_os, status_os) ~ afp_prior_sbrt_cat,
  data = model_overall
)
summary(cox_afp)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ afp_prior_sbrt_cat,
##     data = model_overall)
##
##   n= 77, number of events= 49
##
##                      coef exp(coef) se(coef)     z Pr(>|z|)
## afp_prior_sbrt_cat 0.1009    1.1061   0.2939 0.343    0.731
##
##                    exp(coef) exp(-coef) lower .95 upper .95
## afp_prior_sbrt_cat     1.106      0.904    0.6218     1.968
##
## Concordance= 0.536  (se = 0.041 )
## Likelihood ratio test= 0.12  on 1 df,   p=0.7
## Wald test            = 0.12  on 1 df,   p=0.7
## Score (logrank) test = 0.12  on 1 df,   p=0.7
```

```r
model_overall <- model_overall %>%
  mutate(
    cp_prior_group = relevel(factor(cp_prior_group), ref = "B/C")
  )

cox_cp <- coxph(
  Surv(time_os, status_os) ~ cp_prior_group,
  data = model_overall
)
summary(cox_cp)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ cp_prior_group, data = model_overall)
##
##   n= 77, number of events= 49
##
##                     coef exp(coef) se(coef)      z Pr(>|z|)
## cp_prior_groupA -0.08991   0.91401  0.31165 -0.289    0.773
##
##                 exp(coef) exp(-coef) lower .95 upper .95
## cp_prior_groupA     0.914      1.094    0.4962     1.684
##
## Concordance= 0.547  (se = 0.04 )
## Likelihood ratio test= 0.08  on 1 df,   p=0.8
## Wald test            = 0.08  on 1 df,   p=0.8
## Score (logrank) test = 0.08  on 1 df,   p=0.8
```

```r
model_overall <- model_overall %>%
  mutate(
```

```r
    hcc_size_gt = if_else(hcc_size_sbrt > 3, 1L, 0L)
  )

cox_hccsize <- coxph(
  Surv(time_os, status_os) ~ hcc_size_gt,
  data = model_overall
)
summary(cox_hccsize)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ hcc_size_gt, data = model_overall)
##
##   n= 77, number of events= 49
##
##               coef exp(coef) se(coef)     z Pr(>|z|)
## hcc_size_gt 0.7549    2.1273   0.3429 2.202   0.0277 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##             exp(coef) exp(-coef) lower .95 upper .95
## hcc_size_gt     2.127     0.4701     1.086     4.166
##
## Concordance= 0.579  (se = 0.037 )
## Likelihood ratio test= 5.48  on 1 df,   p=0.02
## Wald test            = 4.85  on 1 df,   p=0.03
## Score (logrank) test = 5.08  on 1 df,   p=0.02
```

```r
model_overall <- model_overall %>%
  mutate(
    eqd2_gt80 = if_else(eqd2 > 80, 1L, 0L)
  )

cox_eqd2 <- coxph(
  Surv(time_os, status_os) ~ eqd2_gt80,
  data = model_overall
)
summary(cox_eqd2)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ eqd2_gt80, data = model_overall)
##
##   n= 77, number of events= 49
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## eqd2_gt80 -0.2058    0.8140   0.2867 -0.718    0.473
##
##           exp(coef) exp(-coef) lower .95 upper .95
## eqd2_gt80     0.814      1.229    0.4641     1.428
##
## Concordance= 0.532  (se = 0.04 )
## Likelihood ratio test= 0.51  on 1 df,   p=0.5
## Wald test            = 0.52  on 1 df,   p=0.5
## Score (logrank) test = 0.52  on 1 df,   p=0.5
```

9.1 Multivariable Cox model

```
model_overall <- model_overall %>%
  mutate(
    Desc_Salv_Consol = relevel(factor(Desc_Salv_Consol), ref = "Salvage"),
    hcc_size_gt      = factor(if_else(hcc_size_sbrt > 3, 1L, 0L)),
    age_gt60         = factor(if_else(age_first_SBRT > 60, 1L, 0L)),
    Milan_criteria   = relevel(factor(Milan_criteria), ref = "No"),
    listed           = factor(listed),
    transplant       = factor(transplant),
    pvtt             = factor(pvtt),
    cp_prior_group   = relevel(factor(cp_prior_group), ref = "B/C"),
    afp_prior_sbrt_cat = factor(if_else(afp_prior_sbrt >= 10, 1L, 0L)),
    eqd2_gt80        = factor(if_else(eqd2 > 80, 1L, 0L))
  )

forced_vars <- c("Desc_Salv_Consol", "hcc_size_gt")
candidates  <- c("age_gt60","Milan_criteria","listed","transplant","pvtt",
                 "afp_prior_sbrt_cat","cp_prior_group","eqd2_gt80")

dat <- model_overall %>%
  select(time_os, status_os, all_of(forced_vars), all_of(candidates)) %>%
  na.omit()

base_f <- as.formula(
  paste("Surv(time_os, status_os) ~", paste(forced_vars, collapse = " + "))
)

full_f <- as.formula(
  paste("Surv(time_os, status_os) ~",
        paste(c(forced_vars, candidates), collapse = " + "))
)

fit_base <- coxph(base_f, data = dat, x = TRUE, y = TRUE)

fit_forward_aic <- step(
  fit_base,
  scope = list(lower = base_f, upper = full_f),
  direction = "forward",
  trace = 1
)
```

```
## Start:  AIC=354.59
## Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt
##
##                       Df    AIC
## + transplant           1 338.34
## + listed               1 350.43
## <none>                   354.59
## + pvtt                 1 354.61
## + age_gt60             1 356.27
## + afp_prior_sbrt_cat   1 356.45
## + Milan_criteria       1 356.47
```

```
## + eqd2_gt80           1 356.47
## + cp_prior_group      1 356.54
##
## Step:  AIC=338.34
## Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + transplant
##
##                      Df    AIC
## + cp_prior_group      1 333.62
## + eqd2_gt80           1 337.07
## <none>                  338.34
## + afp_prior_sbrt_cat  1 338.99
## + pvtt                1 339.44
## + Milan_criteria      1 340.18
## + age_gt60            1 340.30
## + listed              1 340.34
##
## Step:  AIC=333.62
## Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + transplant +
##     cp_prior_group
##
##                      Df    AIC
## <none>                  333.62
## + pvtt                1 334.29
## + afp_prior_sbrt_cat  1 334.82
## + listed              1 335.19
## + eqd2_gt80           1 335.44
## + age_gt60            1 335.50
## + Milan_criteria      1 335.54
```

```r
summary(fit_forward_aic)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + transplant + cp_prior_group, data = dat, x = TRUE,
##     y = TRUE)
##
##   n= 77, number of events= 49
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.2099    1.2335   0.3113  0.674  0.50027
## hcc_size_gt1                     0.5153    1.6742   0.3481  1.480  0.13883
## transplantYes                   -2.1312    0.1187   0.4766 -4.472 7.76e-06 ***
## cp_prior_groupA                 -0.9996    0.3680   0.3723 -2.685  0.00726 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    1.2335     0.8107   0.67009    2.2706
## hcc_size_gt1                       1.6742     0.5973   0.84616    3.3123
## transplantYes                      0.1187     8.4246   0.04664    0.3021
## cp_prior_groupA                    0.3680     2.7171   0.17741    0.7635
##
## Concordance= 0.732  (se = 0.037 )
## Likelihood ratio test= 30.62  on 4 df,   p=4e-06
```

```
## Wald test            = 23.88  on 4 df,   p=8e-05
## Score (logrank) test = 25.88  on 4 df,   p=3e-05
```

```
hr_tbl <- data.frame(
  Variable = rownames(coef(summary(fit_forward_aic))),
  HR       = exp(coef(fit_forward_aic)),
  LCL      = exp(confint(fit_forward_aic)[,1]),
  UCL      = exp(confint(fit_forward_aic)[,2]),
  p_value  = coef(summary(fit_forward_aic))[,"Pr(>|z|)"],
  row.names = NULL
)

hr_tbl
```

```
##                          Variable        HR        LCL       UCL      p_value
## 1 Desc_Salv_ConsolConsolidate/CMV 1.2334958 0.67009481 2.2705920 5.002739e-01
## 2                     hcc_size_gt1 1.6741505 0.84616219 3.3123436 1.388330e-01
## 3                    transplantYes 0.1186998 0.04664301 0.3020738 7.756392e-06
## 4                  cp_prior_groupA 0.3680436 0.17741283 0.7635080 7.259357e-03
```

```
cox.zph(fit_forward_aic)
```

```
##                    chisq df    p
## Desc_Salv_Consol 0.0720  1 0.79
## hcc_size_gt      0.0163  1 0.90
## transplant       0.4473  1 0.50
## cp_prior_group   2.3719  1 0.12
## GLOBAL           2.8192  4 0.59
```

9.2 Manual model building

```
dat <- model_overall %>%
  mutate(
    Desc_Salv_Consol = relevel(factor(Desc_Salv_Consol), ref = "Salvage"),
    hcc_size_gt      = factor(if_else(hcc_size_sbrt > 3, 1L, 0L))
  )

fit_start <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt,
                   data = dat)
summary(fit_start)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt, data = dat)
##
##   n= 77, number of events= 49
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV -0.1219    0.8852   0.2994 -0.407   0.6838
## hcc_size_gt1                     0.7478    2.1124   0.3432  2.179   0.0294 *
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                  exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    0.8852     1.1297    0.4923     1.592
## hcc_size_gt1                       2.1124     0.4734    1.0780     4.140
##
## Concordance= 0.582  (se = 0.042 )
## Likelihood ratio test= 5.65  on 2 df,   p=0.06
## Wald test            = 5.01  on 2 df,   p=0.08
## Score (logrank) test = 5.25  on 2 df,   p=0.07
```

```
AIC(fit_start)
```

```
## [1] 354.5932
```

```
fit_2 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + age_gt60,
               data = dat)
summary(fit_2)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + age_gt60, data = dat)
##
##   n= 77, number of events= 49
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV -0.1502    0.8605   0.3037 -0.495   0.6208
## hcc_size_gt1                     0.7478    2.1124   0.3432  2.179   0.0294 *
## age_gt601                       -0.1857    0.8305   0.3218 -0.577   0.5639
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                  exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    0.8605     1.1621    0.4746     1.560
## hcc_size_gt1                       2.1124     0.4734    1.0780     4.139
## age_gt601                          0.8305     1.2041    0.4420     1.561
##
## Concordance= 0.581  (se = 0.043 )
## Likelihood ratio test= 5.97  on 3 df,   p=0.1
## Wald test            = 5.33  on 3 df,   p=0.1
## Score (logrank) test = 5.57  on 3 df,   p=0.1
```

```
AIC(fit_2)
```

```
## [1] 356.2687
```

```
fit_3 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt,
               data = dat)
summary(fit_3)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt, data = dat)
##
##   n= 77, number of events= 49
##
##                                     coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV -0.03998   0.96080  0.30819 -0.130   0.8968
## hcc_size_gt1                     0.83699   2.30940  0.35213  2.377   0.0175 *
## pvttYes                          0.86019   2.36360  0.55727  1.544   0.1227
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    0.9608     1.0408    0.5252     1.758
## hcc_size_gt1                       2.3094     0.4330    1.1581     4.605
## pvttYes                            2.3636     0.4231    0.7929     7.046
##
## Concordance= 0.571  (se = 0.044 )
## Likelihood ratio test= 7.63  on 3 df,   p=0.05
## Wald test            = 7.04  on 3 df,   p=0.07
## Score (logrank) test = 7.31  on 3 df,   p=0.06
```

```r
AIC(fit_3)
```

```
## [1] 354.6117
```

```r
fit_4 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbrt_cat,
               data = dat)
summary(fit_4)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat, data = dat)
##
##   n= 77, number of events= 49
##
##                                     coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV -0.04039   0.96041  0.30844 -0.131   0.8958
## hcc_size_gt1                     0.83499   2.30479  0.35169  2.374   0.0176 *
## pvttYes                          0.88717   2.42824  0.55970  1.585   0.1130
## afp_prior_sbrt_cat1              0.14894   1.16060  0.30048  0.496   0.6201
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    0.9604     1.0412    0.5247     1.758
## hcc_size_gt1                       2.3048     0.4339    1.1568     4.592
## pvttYes                            2.4282     0.4118    0.8107     7.273
## afp_prior_sbrt_cat1                1.1606     0.8616    0.6440     2.091
##
## Concordance= 0.595  (se = 0.051 )
## Likelihood ratio test= 7.88  on 4 df,   p=0.1
```

```
## Wald test            = 7.38  on 4 df,   p=0.1
## Score (logrank) test = 7.62  on 4 df,   p=0.1
```

```r
AIC(fit_4)
```

```
## [1] 356.3677
```

```r
fit_5 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbrt_cat + M
               data = dat)
summary(fit_5)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat + Milan_criteria,
##     data = dat)
##
##   n= 77, number of events= 49
##
##                                     coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.01115   1.01121  0.33242  0.034    0.973
## hcc_size_gt1                     0.81050   2.24902  0.35651  2.273    0.023 *
## pvttYes                          0.91410   2.49453  0.56363  1.622    0.105
## afp_prior_sbrt_cat1              0.12426   1.13231  0.30536  0.407    0.684
## Milan_criteriaYes               -0.14349   0.86633  0.33609 -0.427    0.669
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    1.0112     0.9889    0.5271     1.940
## hcc_size_gt1                       2.2490     0.4446    1.1182     4.523
## pvttYes                            2.4945     0.4009    0.8265     7.529
## afp_prior_sbrt_cat1                1.1323     0.8832    0.6224     2.060
## Milan_criteriaYes                  0.8663     1.1543    0.4483     1.674
##
## Concordance= 0.595  (se = 0.049 )
## Likelihood ratio test= 8.06  on 5 df,   p=0.2
## Wald test            = 7.61  on 5 df,   p=0.2
## Score (logrank) test = 7.85  on 5 df,   p=0.2
```

```r
AIC(fit_5)
```

```
## [1] 358.1872
```

```r
fit_6 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbrt_cat + l
               data = dat)
summary(fit_6)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat + listed, data = dat)
##
```

```
##   n= 77, number of events= 49
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.1755    1.1919   0.3227  0.544   0.5865
## hcc_size_gt1                     0.7721    2.1643   0.3532  2.186   0.0288 *
## pvttYes                          0.8575    2.3573   0.5574  1.538   0.1240
## afp_prior_sbrt_cat1              0.2489    1.2826   0.3032  0.821   0.4118
## listedYes                       -0.8165    0.4420   0.3296 -2.477   0.0132 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV     1.192     0.8390    0.6333    2.2433
## hcc_size_gt1                        2.164     0.4620    1.0832    4.3244
## pvttYes                             2.357     0.4242    0.7906    7.0292
## afp_prior_sbrt_cat1                 1.283     0.7797    0.7079    2.3237
## listedYes                           0.442     2.2625    0.2316    0.8433
##
## Concordance= 0.619  (se = 0.045 )
## Likelihood ratio test= 14.32  on 5 df,   p=0.01
## Wald test            = 13.2  on 5 df,   p=0.02
## Score (logrank) test = 13.71  on 5 df,   p=0.02
```

```r
AIC(fit_6)
```

```
## [1] 351.9285
```

```r
fit_7 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbrt_cat + l
               data = dat)
summary(fit_7)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat + listed + transplant,
##     data = dat)
##
##   n= 77, number of events= 49
##
##                                     coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.33365   1.39605  0.32207  1.036 0.300229
## hcc_size_gt1                     0.75940   2.13699  0.36473  2.082 0.037335 *
## pvttYes                          0.66262   1.93986  0.56866  1.165 0.243928
## afp_prior_sbrt_cat1              0.39880   1.49004  0.30990  1.287 0.198139
## listedYes                       -0.06781   0.93444  0.36358 -0.187 0.852048
## transplantYes                   -1.58871   0.20419  0.47591 -3.338 0.000843 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    1.3961     0.7163   0.74260    2.6245
## hcc_size_gt1                       2.1370     0.4679   1.04555    4.3678
## pvttYes                            1.9399     0.5155   0.63639    5.9131
## afp_prior_sbrt_cat1                1.4900     0.6711   0.81173    2.7352
```

```
## listedYes                          0.9344     1.0702   0.45822    1.9056
## transplantYes                      0.2042     4.8974   0.08034    0.5189
##
## Concordance= 0.705  (se = 0.041 )
## Likelihood ratio test= 26.45  on 6 df,   p=2e-04
## Wald test            = 20.1  on 6 df,   p=0.003
## Score (logrank) test = 22.69  on 6 df,   p=9e-04
```

```
AIC(fit_7)
```

```
## [1] 341.7939
```

```
fit_8 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbrt_cat + l
               data = dat)
summary(fit_8)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat + listed + transplant +
##     cp_prior_group, data = dat)
##
##   n= 77, number of events= 49
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.3453    1.4124   0.3243  1.065  0.28700
## hcc_size_gt1                     0.7167    2.0477   0.3712  1.931  0.05349 .
## pvttYes                          0.8761    2.4015   0.5819  1.506  0.13219
## afp_prior_sbrt_cat1              0.3430    1.4092   0.3199  1.072  0.28367
## listedYes                       -0.3010    0.7401   0.3678 -0.818  0.41315
## transplantYes                   -2.0998    0.1225   0.5140 -4.085 4.41e-05 ***
## cp_prior_groupA                 -1.1193    0.3265   0.4053 -2.762  0.00575 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    1.4124     0.7080   0.74802    2.6668
## hcc_size_gt1                       2.0477     0.4884   0.98929    4.2384
## pvttYes                            2.4015     0.4164   0.76761    7.5134
## afp_prior_sbrt_cat1                1.4092     0.7096   0.75273    2.6381
## listedYes                          0.7401     1.3512   0.35989    1.5218
## transplantYes                      0.1225     8.1648   0.04472    0.3354
## cp_prior_groupA                    0.3265     3.0627   0.14754    0.7225
##
## Concordance= 0.745  (se = 0.039 )
## Likelihood ratio test= 33.78  on 7 df,   p=2e-05
## Wald test            = 23.84  on 7 df,   p=0.001
## Score (logrank) test = 26.78  on 7 df,   p=4e-04
```

```
AIC(fit_8)
```

```
## [1] 336.459
```

```r
fit_9 <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbrt_cat + l
               data = dat)
summary(fit_9)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat + listed + transplant +
##     cp_prior_group + eqd2_gt80, data = dat)
##
##   n= 77, number of events= 49
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.34744   1.41544  0.32490  1.069   0.2849
## hcc_size_gt1                     0.69315   2.00001  0.38464  1.802   0.0715 .
## pvttYes                          0.88242   2.41674  0.58302  1.514   0.1301
## afp_prior_sbrt_cat1              0.33134   1.39284  0.32346  1.024   0.3057
## listedYes                       -0.28723   0.75034  0.37426 -0.767   0.4428
## transplantYes                   -2.11229   0.12096  0.51748 -4.082 4.47e-05 ***
## cp_prior_groupA                 -1.04862   0.35042  0.50176 -2.090   0.0366 *
## eqd2_gt801                      -0.09946   0.90533  0.42061 -0.236   0.8131
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    1.4154     0.7065   0.74875    2.6757
## hcc_size_gt1                       2.0000     0.5000   0.94108    4.2505
## pvttYes                            2.4167     0.4138   0.77083    7.5771
## afp_prior_sbrt_cat1                1.3928     0.7180   0.73888    2.6256
## listedYes                          0.7503     1.3327   0.36032    1.5625
## transplantYes                      0.1210     8.2671   0.04387    0.3335
## cp_prior_groupA                    0.3504     2.8537   0.13107    0.9369
## eqd2_gt801                         0.9053     1.1046   0.39699    2.0646
##
## Concordance= 0.74  (se = 0.04 )
## Likelihood ratio test= 33.84  on 8 df,   p=4e-05
## Wald test            = 23.91  on 8 df,   p=0.002
## Score (logrank) test = 26.85  on 8 df,   p=8e-04
```

```r
AIC(fit_9)
```

```
## [1] 338.4037
```

```r
final_model_os <- coxph(Surv(time_os, status_os) ~ Desc_Salv_Consol + hcc_size_gt + pvtt + afp_prior_sbr
                        data = dat)
summary(final_model_os)
```

```
## Call:
## coxph(formula = Surv(time_os, status_os) ~ Desc_Salv_Consol +
##     hcc_size_gt + pvtt + afp_prior_sbrt_cat + listed + transplant +
##     cp_prior_group, data = dat)
##
##   n= 77, number of events= 49
```

```
##
##                                    coef exp(coef) se(coef)      z Pr(>|z|)
## Desc_Salv_ConsolConsolidate/CMV  0.3453    1.4124   0.3243  1.065  0.28700
## hcc_size_gt1                     0.7167    2.0477   0.3712  1.931  0.05349 .
## pvttYes                          0.8761    2.4015   0.5819  1.506  0.13219
## afp_prior_sbrt_cat1              0.3430    1.4092   0.3199  1.072  0.28367
## listedYes                       -0.3010    0.7401   0.3678 -0.818  0.41315
## transplantYes                   -2.0998    0.1225   0.5140 -4.085 4.41e-05 ***
## cp_prior_groupA                 -1.1193    0.3265   0.4053 -2.762  0.00575 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                                 exp(coef) exp(-coef) lower .95 upper .95
## Desc_Salv_ConsolConsolidate/CMV    1.4124     0.7080   0.74802    2.6668
## hcc_size_gt1                       2.0477     0.4884   0.98929    4.2384
## pvttYes                            2.4015     0.4164   0.76761    7.5134
## afp_prior_sbrt_cat1                1.4092     0.7096   0.75273    2.6381
## listedYes                          0.7401     1.3512   0.35989    1.5218
## transplantYes                      0.1225     8.1648   0.04472    0.3354
## cp_prior_groupA                    0.3265     3.0627   0.14754    0.7225
##
## Concordance= 0.745  (se = 0.039 )
## Likelihood ratio test= 33.78  on 7 df,   p=2e-05
## Wald test            = 23.84  on 7 df,   p=0.001
## Score (logrank) test = 26.78  on 7 df,   p=4e-04
```

```
AIC(final_model_os)
```

```
## [1] 336.459
```

10 Additional analyses

10.1 Recurrent vs residual tumors

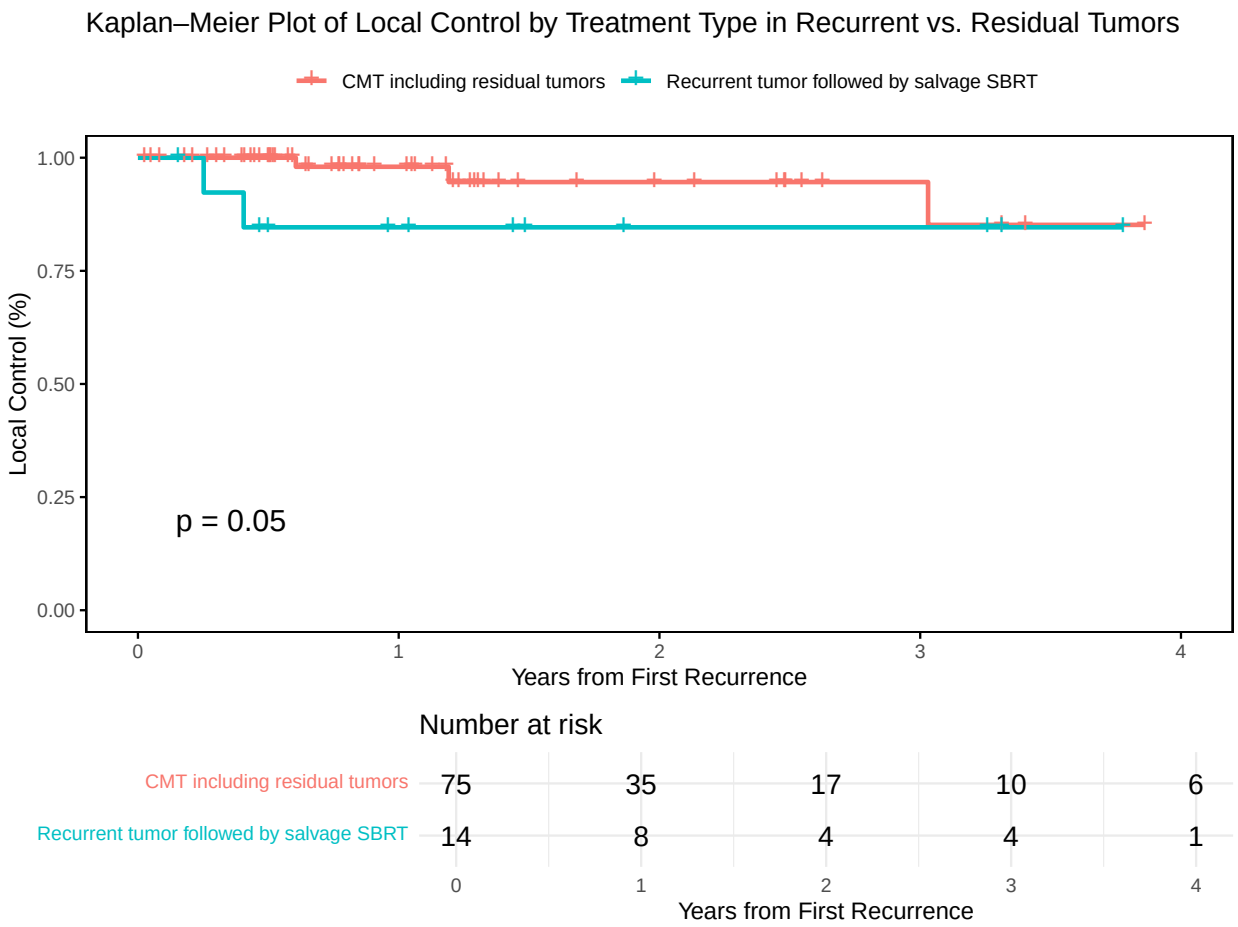
```
model_lc <- model_set1 %>%
  filter(residual_reccurence %in% c("Residual", "Recurrence", "Not included"))

table9_summary <- model_lc %>%
  group_by(residual_reccurence) %>%
  summarise(N = n(), .groups = "drop") %>%
  mutate(Percent = round(100 * N / sum(N), 1))

table9_summary
```

```
## # A tibble: 3 x 3
##   residual_reccurence     N Percent
##   <fct>               <int>   <dbl>
## 1 Residual               42    47.2
## 2 Recurrence             14    15.7
## 3 Not included           33    37.1
```

```r
model_km <- model_set1 %>%
  mutate(
    group_recur = if_else(residual_reccurence == "Recurrence",
                          "Recurrent tumor followed by salvage SBRT",
                          "CMT including residual tumors")
  )

fit_km <- survfit(Surv(time_lc_yr, resp_fail) ~ group_recur, data = model_km)

lc_summary <- summary(fit_km, times = c(1:4))
km_table <- data.frame(
  Group   = rep(levels(factor(model_km$group_recur)), each = length(lc_summary$time) / 2),
  Time    = lc_summary$time,
  LC_rate = round(lc_summary$surv * 100, 1)
)

print(km_table)
```

```
##                                      Group Time LC_rate
## 1            CMT including residual tumors    1    98.0
## 2            CMT including residual tumors    2    94.6
## 3            CMT including residual tumors    3    94.6
## 4            CMT including residual tumors    4    85.2
## 5 Recurrent tumor followed by salvage SBRT    1    84.6
## 6 Recurrent tumor followed by salvage SBRT    2    84.6
## 7 Recurrent tumor followed by salvage SBRT    3    84.6
## 8 Recurrent tumor followed by salvage SBRT    4    84.6
```

```r
plot_recurrent_group <- ggsurvplot(
  fit_km,
  data = model_km,
  risk.table = TRUE,
  pval = sprintf("p = %.2f", surv_pvalue(fit_km, data = model_km)$pval),
  conf.int = FALSE,
  xlab = "Years from First Recurrence",
  ylab = "Local Control (%)",
  title = "Kaplan-Meier Plot of Local Control by Treatment Type in Recurrent vs. Residual Tumors",
  legend.title = "",
  legend.labs = c("CMT including residual tumors", "Recurrent tumor followed by salvage SBRT"),
  break.time.by = 1,
  xlim = c(0, 4),
  ggtheme = theme_minimal()
)

plot_recurrent_group$plot <- plot_recurrent_group$plot +
  scale_x_continuous(limits = c(0, 4), breaks = 0:4) +
  theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid = element_blank()
  )

print(plot_recurrent_group)
```



10.2 Recurrent vs residual probability table and RMST

```
fit_km <- survfit(
  Surv(time_lc_yr, resp_fail) ~ group_recur,
  data = model_km,
  conf.type = "log-log"
)

lc_summary <- summary(fit_km, times = c(1, 2, 3, 4))

lc_df <- data.frame(
  Group    = lc_summary$strata,
  Time     = lc_summary$time,
  N_Risk   = lc_summary$n.risk,
  N_Event  = lc_summary$n.event,
  Survival = round(lc_summary$surv  * 100, 1),
  LCL      = round(lc_summary$lower * 100, 1),
  UCL      = round(lc_summary$upper * 100, 1),
  row.names = NULL
)
```

```r
lc_df$Group <- sub("^.*=", "", lc_df$Group)
lc_df
```

```
##                                      Group Time N_Risk N_Event Survival  LCL
## 1            CMT including residual tumors    1     35       1     98.0 86.6
## 2            CMT including residual tumors    2     17       1     94.6 79.3
## 3            CMT including residual tumors    3     10       0     94.6 79.3
## 4            CMT including residual tumors    4      6       1     85.2 52.9
## 5 Recurrent tumor followed by salvage SBRT    1      8       2     84.6 51.2
## 6 Recurrent tumor followed by salvage SBRT    2      4       0     84.6 51.2
## 7 Recurrent tumor followed by salvage SBRT    3      4       0     84.6 51.2
## 8 Recurrent tumor followed by salvage SBRT    4      1       0     84.6 51.2
##    UCL
## 1 99.7
## 2 98.7
## 3 98.7
## 4 96.0
## 5 95.9
## 6 95.9
## 7 95.9
## 8 95.9
```

```r
summary(fit_km)$table
```

```
##                                                      records n.max n.start
## group_recur=CMT including residual tumors                 75    75      75
## group_recur=Recurrent tumor followed by salvage SBRT      14    14      14
##                                                      events    rmean se(rmean)
## group_recur=CMT including residual tumors                 3 6.517180 0.4394967
## group_recur=Recurrent tumor followed by salvage SBRT      3 3.871022 0.4189990
##                                                        median 0.95LCL 0.95UCL
## group_recur=CMT including residual tumors                  NA      NA      NA
## group_recur=Recurrent tumor followed by salvage SBRT 4.515068      NA      NA
```

```r
max_by_arm <- with(model_km,
                   tapply(time_lc_yr, group_recur, max, na.rm = TRUE))
max_by_arm
```

```
##            CMT including residual tumors
##                                 7.254795
## Recurrent tumor followed by salvage SBRT
##                                 4.515068
```

```r
tau <- 4.5

rmst_tbl <- summary(fit_km, rmean = tau)$table |>
  as.data.frame(check.names = FALSE) |>
  tibble::rownames_to_column("group") |>
  dplyr::transmute(
    group     = sub("^.*=", "", group),
    tau       = tau,
```

```
    rmst      = rmean,
    se_rmst   = `se(rmean)`,
    lower_95  = rmst - 1.96 * se_rmst,
    upper_95  = rmst + 1.96 * se_rmst
  )
rmst_tbl
```

```
##                                      group tau     rmst   se_rmst lower_95
## 1            CMT including residual tumors 4.5 4.171235 0.1833496 3.811870
## 2 Recurrent tumor followed by salvage SBRT 4.5 3.858272 0.4174914 3.039989
##   upper_95
## 1 4.530600
## 2 4.676555
```

10.3 Tau selection before heavy censoring for recurrent vs residual

```
cutoff_tau_before_heavy_censoring <- function(fit, by = 0.25, risk_threshold = 0.10) {
  s_all <- summary(fit)
  t_max <- min(tapply(s_all$time, s_all$strata, max, na.rm = TRUE), na.rm = TRUE)
  times <- seq(0, t_max, by = by)

  s <- summary(fit, times = times)

  df <- tibble(
    time   = s$time,
    strata = s$strata,
    n_risk = s$n.risk
  )

  n0 <- df %>%
    group_by(strata) %>%
    summarise(N0 = max(n_risk, na.rm = TRUE), .groups = "drop")

  df2 <- df %>%
    inner_join(n0, by = "strata") %>%
    mutate(frac_at_risk = n_risk / N0)

  limiting <- df2 %>%
    group_by(time) %>%
    summarise(min_frac_at_risk = min(frac_at_risk, na.rm = TRUE), .groups = "drop")

  tau_suggested <- limiting %>%
    filter(min_frac_at_risk >= risk_threshold) %>%
    summarise(tau = max(time, na.rm = TRUE)) %>%
    pull(tau)

  list(
    tau_suggested       = tau_suggested,
    limiting_support    = limiting,
    per_time_support    = df2,
    t_max_shortest      = t_max,
```

```r
    risk_threshold_used = risk_threshold,
    grid_by             = by
  )
}

out_tau_lc <- cutoff_tau_before_heavy_censoring(fit_km, by = 0.25, risk_threshold = 0.10)
tau_lc <- out_tau_lc$tau_suggested
tau_lc
```

```
## [1] 3
```

```r
rmst_lc_tbl <- summary(fit_km, rmean = tau_lc)$table %>%
  as.data.frame() %>%
  rownames_to_column("group") %>%
  transmute(
    group,
    n        = records,
    events   = events,
    tau      = tau_lc,
    rmst     = rmean,
    se_rmst  = `se(rmean)`,
    lower_95 = rmean - 1.96 * se_rmst,
    upper_95 = rmean + 1.96 * se_rmst
  ) %>%
  mutate(across(c(tau, rmst, se_rmst, lower_95, upper_95), ~ round(., 2)))

rmst_lc_tbl
```

```
##                                                   group  n events tau rmst
## 1            group_recur=CMT including residual tumors 75      3   3 2.89
## 2 group_recur=Recurrent tumor followed by salvage SBRT 14      3   3 2.59
##   se_rmst lower_95 upper_95
## 1    0.08     2.74     3.04
## 2    0.27     2.06     3.11
```

10.4 CMT diagnostic imaging subgroup

```r
model_set1 <- model_set1 %>%
  mutate(
    cmt_diag_test_bin = case_when(
      cmt_diag_test == "No"  ~ 0,
      cmt_diag_test == "Yes" ~ 1,
      cmt_diag_test == "Not included" ~ NA_real_
    )
  )

table(model_set1$cmt_diag_test, model_set1$cmt_diag_test_bin, useNA = "ifany")
```

```
##
##                 0  1 <NA>
```

```
##   No           12  0    0
##   Yes           0 21    0
##   Not included  0  0   56
```

```
cmt_subset <- model_set1 %>%
  filter(Desc_Salv_Consol == "Consolidate/CMV") %>%
  mutate(
    cmt_diag_test_bin = dplyr::case_when(
      cmt_diag_test == "No"  ~ 0,
      cmt_diag_test == "Yes" ~ 1,
      TRUE ~ NA_real_
    ),
    diag_group = if_else(cmt_diag_test_bin == 1,
                         "Diagnostic imaging between tx",
                         "No diagnostic imaging between tx")
  ) %>%
  filter(!is.na(cmt_diag_test_bin))

table(cmt_subset$diag_group, useNA = "ifany")
```

```
##
##    Diagnostic imaging between tx No diagnostic imaging between tx
##                               20                               12
```

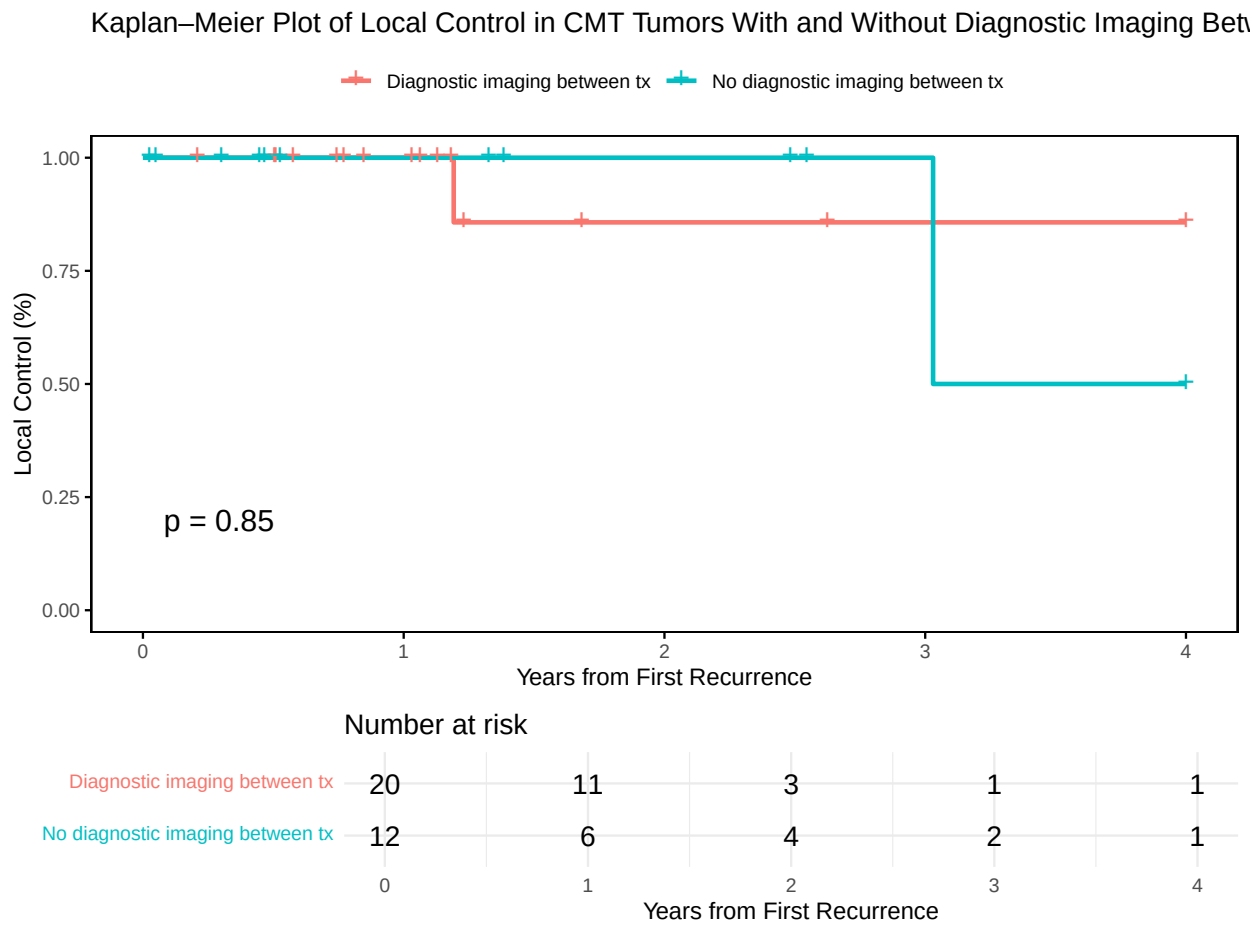
10.5 CMT diagnostic imaging KM and RMST

```
fit_km_cmt <- survfit(
  Surv(time_lc_yr, resp_fail) ~ diag_group,
  data = cmt_subset,
  conf.type = "log-log"
)

lc_summary <- summary(fit_km_cmt, times = c(1, 2, 3, 4))

lc_df <- data.frame(
  Group    = sub("^.*=", "", lc_summary$strata),
  Time     = lc_summary$time,
  N_Risk   = lc_summary$n.risk,
  N_Event  = lc_summary$n.event,
  Survival = round(lc_summary$surv  * 100, 1),
  LCL      = round(lc_summary$lower * 100, 1),
  UCL      = round(lc_summary$upper * 100, 1),
  row.names = NULL
)
lc_df
```

```
##                              Group Time N_Risk N_Event Survival  LCL  UCL
## 1    Diagnostic imaging between tx    1     11       0    100.0   NA   NA
## 2    Diagnostic imaging between tx    2      3       1     85.7 33.4 97.9
## 3    Diagnostic imaging between tx    3      1       0     85.7 33.4 97.9
## 4    Diagnostic imaging between tx    4      1       0     85.7 33.4 97.9
```

```
## 5 No diagnostic imaging between tx    1      6       0    100.0   NA   NA
## 6 No diagnostic imaging between tx    2      4       0    100.0   NA   NA
## 7 No diagnostic imaging between tx    3      2       0    100.0   NA   NA
## 8 No diagnostic imaging between tx    4      1       1     50.0  0.6 91.0
```

```r
cmt_trunc <- cmt_subset %>%
  mutate(
    time4   = pmin(time_lc_yr, 4),
    status4 = ifelse(resp_fail == 1 & time_lc_yr <= 4, 1, 0)
  )

fit_km_cmt_trim <- survfit(Surv(time4, status4) ~ diag_group,
                           data = cmt_trunc, conf.type = "log-log")

km <- ggsurvplot(
  fit_km_cmt_trim,
  data = cmt_trunc,
  conf.int   = FALSE,
  pval       = TRUE,
  risk.table = TRUE,
  xlim       = c(0, 4),
  title = "Kaplan-Meier Plot of Local Control in CMT Tumors With and Without Diagnostic Imaging Between
  break.time.by = 1,
  ggtheme    = theme_minimal(),
  xlab       = "Years from First Recurrence",
  ylab       = "Local Control (%)",
  legend.title = "",
  legend.labs  = c("Diagnostic imaging between tx", "No diagnostic imaging between tx")
)

km$plot <- km$plot +
  scale_x_continuous(limits = c(0, 4), breaks = 0:4) +
  theme(
    panel.border     = element_rect(color = "black", fill = NA, linewidth = 0.8),
    panel.background = element_blank(),
    panel.grid       = element_blank()
  )

print(km)
```




```r
tapply(cmt_subset$time_lc_yr, cmt_subset$diag_group, max)
```

```
##    Diagnostic imaging between tx No diagnostic imaging between tx
##                         7.254795                         6.189041
```

```r
tau <- min(
  tapply(cmt_subset$time_lc_yr, cmt_subset$diag_group, max)
)
tau
```

```
## [1] 6.189041
```

```r
cmt_subset$diag_bin <- ifelse(
  cmt_subset$diag_group == "Diagnostic imaging between tx", 1, 0
)

rmst_tbl <- summary(fit_km_cmt, rmean = tau)$table |>
  as.data.frame(check.names = FALSE) |>
  tibble::rownames_to_column("group") |>
  dplyr::transmute(
    group     = sub("^.*=", "", group),
    tau       = tau,
```

```
    rmst      = rmean,
    se_rmst   = `se(rmean)`,
    lower_95  = rmst - 1.96 * se_rmst,
    upper_95  = rmst + 1.96 * se_rmst
  )
rmst_tbl
```

```
##                              group      tau     rmst   se_rmst lower_95
## 1    Diagnostic imaging between tx 6.189041 5.475147 0.6609377 4.179709
## 2 No diagnostic imaging between tx 6.189041 4.609589 1.1168413 2.420580
##   upper_95
## 1 6.770585
## 2 6.798598
```

10.6 Alternative tau selection for CMT subgroup

```
km_df <- broom::tidy(fit_km_cmt) %>%
  group_by(strata) %>%
  mutate(risk_prop = n.risk / max(n.risk)) %>%
  summarise(
    tau_suggested = max(time[risk_prop >= 0.10]),
    .groups = "drop"
  )

tau_cmt <- min(km_df$tau_suggested)
tau_cmt
```

```
## [1] 2.624658
```

```
rmst_cmt_tbl <- summary(fit_km_cmt, rmean = tau_cmt)$table %>%
  as.data.frame() %>%
  tibble::rownames_to_column("group") %>%
  transmute(
    group,
    n        = records,
    events   = events,
    tau      = tau_cmt,
    rmst     = rmean,
    se_rmst  = `se(rmean)`,
    lower_95 = rmean - 1.96 * se_rmst,
    upper_95 = rmean + 1.96 * se_rmst
  ) %>%
  mutate(across(c(tau, rmst, se_rmst, lower_95, upper_95), ~ round(., 2)))

rmst_cmt_tbl
```

```
##                                         group  n events  tau rmst se_rmst
## 1    diag_group=Diagnostic imaging between tx 20      1 2.62 2.42    0.19
## 2 diag_group=No diagnostic imaging between tx 12      1 2.62 2.62    0.00
##   lower_95 upper_95
## 1     2.05     2.79
## 2     2.62     2.62
```

10.7 Table 10 interval summary

```
model_set1 <- model_set1 %>%
  mutate(
    sbrt_end = if_else(
      unique_id == 77,
      as.Date("2018-10-19"),
      sbrt_end
    )
  )

model_set1 %>%
  filter(unique_id == 77) %>%
  select(unique_id, date_last_LDT, sbrt_end)
```

```
## # A tibble: 1 x 3
##   unique_id date_last_LDT sbrt_end
##       <dbl> <date>        <date>
## 1        77 2018-08-14    2018-10-19
```

```
model_set1$ldt_to_sbrt_interval_months <- time_length(
  interval(model_set1$date_last_LDT, model_set1$sbrt_end),
  unit = "months"
)

group_summary <- model_set1 %>%
  group_by(Desc_Salv_Consol) %>%
  summarise(
    Median = round(median(ldt_to_sbrt_interval_months, na.rm = TRUE), 1),
    Q1 = round(quantile(ldt_to_sbrt_interval_months, 0.25, na.rm = TRUE), 1),
    Q3 = round(quantile(ldt_to_sbrt_interval_months, 0.75, na.rm = TRUE), 1),
    Range_Low = round(min(ldt_to_sbrt_interval_months, na.rm = TRUE), 1),
    Range_High = round(max(ldt_to_sbrt_interval_months, na.rm = TRUE), 1),
    N = sum(!is.na(ldt_to_sbrt_interval_months)),
    .groups = "drop"
  ) %>%
  mutate(
    IQR = paste0("(", Q1, "-", Q3, ")"),
    Range = paste0("(", Range_Low, "-", Range_High, ")"),
    p_value = NA
  ) %>%
  select(Group = Desc_Salv_Consol, Median, IQR, Range, N, p_value)

total_summary <- model_set1 %>%
  summarise(
    Median = round(median(ldt_to_sbrt_interval_months, na.rm = TRUE), 1),
    Q1 = round(quantile(ldt_to_sbrt_interval_months, 0.25, na.rm = TRUE), 1),
    Q3 = round(quantile(ldt_to_sbrt_interval_months, 0.75, na.rm = TRUE), 1),
    Range_Low = round(min(ldt_to_sbrt_interval_months, na.rm = TRUE), 1),
    Range_High = round(max(ldt_to_sbrt_interval_months, na.rm = TRUE), 1),
    N = sum(!is.na(ldt_to_sbrt_interval_months))
  ) %>%
```

```r
  mutate(
    Group = "Total",
    IQR = paste0("(", Q1, "-", Q3, ")"),
    Range = paste0("(", Range_Low, "-", Range_High, ")"),
    p_value = round(wilcox.test(ldt_to_sbrt_interval_months ~ Desc_Salv_Consol, data = model_set1)$p.val
  ) %>%
  select(Group, Median, IQR, Range, N, p_value)

final_summary <- bind_rows(group_summary, total_summary)
final_summary
```

```
## # A tibble: 3 x 6
##   Group           Median IQR       Range         N p_value
##   <chr>            <dbl> <chr>     <chr>     <int>   <dbl>
## 1 Salvage            4.3 (3.2-9.4) (1.4-30)     57      NA
## 2 Consolidate/CMV    2.1 (1.8-2.8) (1.1-5.3)    32      NA
## 3 Total              3.3 (2.1-7.9) (1.1-30)     89       0
```